The Limiting Distributions of Chi-square Statistic in High-Dimensional Contingency Tables

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Abstract

Pearson's chi-squared statistic is a cornerstone of statistical inference, yet the validity of its familiar distributional approximations becomes subtle when the number of cells increases with the sample size. We establish, to our knowledge, the first general necessary and sufficient conditions for its chi-squared and normal approximations in multinomial models with possibly diverging dimensions and sample-size-dependent, nonuniform cell probabilities. The results distinguish the effects of sparsity, probability imbalance, and variance inflation, and characterize both the conventional chi-squared approximation and a centered approximation with variance-matched degrees of freedom. In the equiprobable case, the chi-squared approximation holds precisely when $n^2/d \to \infty$, and asymptotic normality additionally requires $d \to \infty$. More generally, the theory permits accurate approximations even when empty cells dominate, while identifying configurations in which classical sufficient conditions are unnecessary. A cross-sample version admits a particularly transparent characterization through the expected number of cells containing at least two observations. The theory also yields necessary and sufficient conditions for normal approximations of the log likelihood ratio and Hellinger statistics under the stated side conditions, and extends to two-way tables with estimated marginal probabilities. Matching the mean and variance provides a unified chi-squared approximation across low- and high-dimensional regimes, approaching a normal approximation as the adjusted degrees of freedom diverge. Simulations support the theory and illustrate the improved calibration obtained by the moment-matched approximations in sparse tables.

Keywords: Central limit theorem; Chi-squared approximation; Goodness-of-fit; Hoeffding's decomposition; Orthogonal projection; Poissonization

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1 Introduction

Pearson's chi-squared statistic is one of the foundational objects of statistical inference. Since its introduction by Pearson (1900), its chi-squared approximation has provided a standard basis for analyzing categorical data. The classical theory assumes a fixed number of cells and sufficiently large expected counts. In modern applications, however, the number of categories or discretization bins may increase with the sample size, while many expected counts remain small or vanish. This raises a basic distributional question: under precisely what conditions do chi-squared and normal approximations remain valid in a high-dimensional multinomial model? Answering this question requires identifying both the conditions that guarantee an approximation and the obstructions that make it fail.

Let $X_{1:d} = (X_1, \ldots, X_d)^\top \sim \text{multinomial}(n, \theta_1, \ldots, \theta_d)$, where $d = d_n \geq 2$, $\theta_j = \theta_{j,n} > 0$, and $\sum_{j=1}^d \theta_j = 1$. Pearson's statistic is

$$\chi^2 = \sum_{j=1}^{d} \frac{(X_j - n\theta_j)^2}{n\theta_j}.$$

For fixed d and fixed positive cell probabilities, χ^2 converges in distribution to χ^2_{d-1} as $n \to \infty$. When d also diverges, the reference distribution itself changes with n, and the problem is to determine whether the approximation error, measured by the supremum distance between the distribution functions, tends to zero. A normal approximation introduces a further question about the appropriate centering and variance. Neither approximation is determined by the dimension and sample size alone when the cell probabilities are nonuniform.

Traditional guidelines typically require all expected counts to exceed a fixed threshold, or permit only a limited fraction of small expected counts (Cochran, 1952, 1954; Agresti, 2012). Such rules constrain individual cells and can impose sample size requirements of order d. They do not describe the precise asymptotic boundary of a valid approximation. In particular, they cannot explain why a chi-squared or normal approximation may remain accurate in a table with an overwhelming fraction of empty cells, or why two tables with the same dimension and sample size can have different limiting behavior.

There is a substantial literature on distributional approximations for multinomial statistics. Holst (1972) and Morris (1975) established central limit theorems under regularity conditions on expected counts and cell probabilities. **?** identified Gaussian, Poisson, and degenerate limits in important regimes, including a sharp description for equiprobable cells. Xu et al. (2019) developed a high-dimensional approximation theory under Lyapunov-type conditions and studied a diagonal-removed version of Pearson's statistic. **?** obtained further sufficient conditions for asymptotic normality in sparse multinomial models. These results show that accurate approximations extend well beyond classical fixed-dimensional settings. A general characterization of their validity for sample-size-dependent, nonuniform cell probabilities, including necessity as well as sufficiency, has nevertheless remained unavailable. The value of identifying exact approximation boundaries is also evident in the related study of Wilks' phenomenon for Gaussian likelihood ratio statistics (He et al., 2021).

This paper establishes, to our knowledge, the first general necessary and sufficient conditions for chi-squared and normal approximations of Pearson's statistic in multinomial triangular arrays with arbitrary positive cell probabilities. Our results characterize the centered chi-squared approximation with adjusted degrees of freedom and the normal approximation, and identify the validity boundary of the conventional χ^2_{d-1} approximation under its variance-consistency side condition. The theory recovers the classical fixed-dimensional limit and accommodates high-dimensional regimes with highly unequal probabilities and vanishing expected counts. It also identifies which familiar sufficient conditions can be relaxed and which features of the cell probabilities genuinely obstruct an approximation.

A central issue is that matching the mean does not ensure matching the variance. Although $\mathbb{E}[\chi^2] = d - 1$

for every multinomial model, its exact variance is

$$V = 2(d-1)(1-1/n) + \frac{1}{n}\left(\sum_{j=1}^{d} \theta_j^{-1} - d^2\right).$$

The second term reflects probability imbalance and can be comparable to, or much larger than, the conventional chi-squared variance $2(d-1)$. Under the variance-consistency condition $\sum_{j=1}^{d} \theta_j^{-1} - d^2 = o(nd)$, we prove that the conventional χ^2_{d-1} approximation holds if and only if

$$\frac{1}{d}\sum_{j=1}^{d} \frac{1}{1+\sqrt{d}\, n\theta_j} \longrightarrow 0.$$

This averaged condition replaces a restriction on the smallest expected count with a criterion involving the entire probability array. Our more general results also cover configurations in which the imbalance contribution to the variance is nonnegligible.

For equiprobable cells, $\theta_j = 1/d$, the criterion reduces to $n^2/d \to \infty$; asymptotic normality holds if and only if both $d \to \infty$ and $n^2/d \to \infty$. Thus, in the regime $n \ll d \ll n^2$, both approximations remain valid even though every expected count n/d tends to zero. These conclusions agree with the earlier equiprobable theory of **?**; the advance here is the characterization for general probability arrays. Indeed, our examples show that nonuniform models can be asymptotically normal even when $d \gg n^2$, so the equiprobable boundary cannot serve as a universal dimension threshold.

The variance formula also motivates an approximation that matches both moments. With $d^* = V/2$, we approximate the centered statistic $\chi^2 - (d-1)$ by the centered chi-squared variable $\chi^2_{d^*} - d^*$, and give necessary and sufficient conditions for its validity. This construction distinguishes the mean $d-1$ from the degrees of freedom needed to match the variance. It connects the fixed-dimensional chi-squared limit to the high-dimensional normal approximation: when $d^* \to \infty$, the standardized reference distribution converges to the standard normal law. Consequently, the theory provides a common approximation framework across dimensions while retaining the variance contribution of unequal cell probabilities.

We also characterize a cross-sample version of Pearson's statistic, obtained by removing its linear component in the Hoeffding decomposition. When $d \to \infty$, its centered chi-squared approximation holds if and only if the expected number of cells containing at least two observations diverges in every deterministic subset occupying a nonvanishing fraction of all cells. This occupancy interpretation explains how extensive sparsity can coexist with an accurate approximation. The same principle extends to two-way contingency tables with estimated marginal probabilities. With diverging degrees of freedom, vanishing fractions of empty rows and columns, together with the occupancy condition, are necessary and sufficient for the cross-sample statistic. The corresponding characterization for the classical statistic additionally requires the variance side condition stated in the main results. These two-way results provide an application of the distributional theory to independence testing.

The analysis further yields necessary and sufficient conditions for normal approximations of the log likelihood ratio and Hellinger statistics in the variance regimes and under the side conditions specified in our theorems. These results include examples with vanishing expected counts or a dominant cell probability, beyond familiar sufficient conditions. The proofs combine Poissonization, orthogonal projection, and Hoeffding decompositions to express the relevant Lindeberg-type conditions in terms of the multinomial probabilities. Simulations illustrate the resulting approximation accuracy and show how matching the mean and variance improves calibration in sparse and unbalanced tables.

The paper develops the chi-squared approximation theory and its two-way-table applications first, followed by the log likelihood ratio and Hellinger results and numerical studies. Throughout the paper, $[d] = \{1, \ldots, d\}$ and $a_{1:d} = (a_1, \ldots, a_d)^\top$ for any positive integer d and real numbers a_j.

2 Chi-squared statistics

2.1 Test of independence

We consider the test of independence of the row and column effects in a large two-way contingency table. Let $X = (X_{j,k})_{d_1 \times d_2}$ be the data matrix with $X_{j,k}$ being the count for the cell in the j-th row and k-th column. The row, column and grand totals of the matrix X are given by

$$R_j = \sum_{k=1}^{d_2} X_{j,k}, \quad C_k = \sum_{j=1}^{d_1} X_{j,k}, \quad n = \sum_{j=1}^{d_1} R_j = \sum_{k=1}^{d_2} C_k. \tag{1}$$

Formally, we state the null hypothesis in the test of independence as

$$H_0 : \ X = (X_{j,k}, j \in [d_1], k \in [d_2]) \sim \text{multinomial}(n, p_j q_k, j \in [d_1], k \in [d_2]), \tag{2}$$

which means the independence of row and column effects, where $p_1, \ldots, p_{d_1}$ and $q_1, \ldots, q_{d_2}$ are treated as completely unknown parameters satisfying $p_j > 0$, $\sum_{j=1}^{d_1} p_j = 1$, $q_k > 0$, and $\sum_{k=1}^{d_2} q_k = 1$. Throughout the section, n and (d_1, d_2) are treated as deterministic and $d_1 \vee d_2$ are allowed to diverge as $n \to \infty$.

The classic chi-squared test of independence rejects the null hypothesis H_0 in (2) when the following statistic exceeds a certain threshold:

$$\chi^2_{\text{indep}} = \sum_{R_j > 0} \sum_{C_k > 0} \frac{(X_{j,k} - R_j C_k / n)^2}{R_j C_k / n} = \sum_{R_j > 0} \sum_{C_k > 0} \frac{X_{j,k}^2}{R_j C_k / n} - n. \tag{3}$$

The typical practice is to reject H_0 when $\chi^2_{\text{indep}} > t_\alpha$, where t_α is the $(1-\alpha)$-th quantile of the statistic determined by the hypergeometric distribution for $d_1 = d_2 = 2$, known as Fisher's exact test, and approximately determined by the χ^2_ν distribution with degrees of freedom $\nu = (d_1 - 1)(d_2 - 1)$ for more general d_1 and d_2. However, in the case where ν is allowed to diverge as $n \to \infty$, the validity of the χ^2_ν approximation is unclear in the existing literature and nontrivial as our analysis attests.

To best present our results for the χ^2_ν approximation of the above test statistic, we shall first consider the following cross-sample version of it,

$$\chi^2_{\text{CS-indep}} = \sum_{R_j > 0} \sum_{C_k > 0} \frac{X_{j,k}(X_{j,k} - 1)}{R_j C_k / n} - \frac{(n - d_1')(n - d_2')}{n - 1} \tag{4}$$

with $d_1' = \#\{j : R_j > 0\}$ and $d_2' = \#\{k : C_k > 0\}$. This test statistic for independence is motivated by the works of Bai and Saranadasa (1996) on Hotelling's T^2 test and Xu et al. (2019) on Pearson's goodness-of-fit test as we will discuss in Section 2.3 below. Write $X_{j,k} = \sum_{i=1}^n B_{i,j,k}$ with Bernoulli variables $B_{i,j,k}$. We call (4) the cross-sample version of the classic chi-squared statistic (3) because $X_{j,k}(X_{j,k} - 1) = \sum_{1 \le i \ne i' \le n} B_{i,j,k} B_{i'j,k}$. In addition, (4) is conditionally centered under H_0:

$$\mathbb{E}\left[\chi^2_{\text{CS-indep}} \middle| R_{1:d_1}, C_{1:d_2}\right] = 0.$$

The following theorem provides necessary and sufficient conditions for the χ^2_ν approximation of the sampling distribution of this cross-sample test statistic in two-way contingency tables.

Theorem 2.1. *Suppose (2) holds with $d_1 \wedge d_2 \ge 2$. Then, the chi-squared approximation*

$$\sup_{-\infty \le t \le \infty} \left| \mathbb{P}\left\{\chi^2_{\text{CS-indep}} \le t\right\} - \mathbb{P}\left\{\chi^2_\nu - \nu \le t\right\} \right| \to 0 \tag{5}$$

holds with $\nu = (d_1 - 1)(d_2 - 1)$ *for the cross-sample chi-squared statistic in* (4) *if and only if*

$$\frac{1}{d_1 d_2} \sum_{j=1}^{d_1} \sum_{k=1}^{d_2} \left(\frac{1}{1 + np_j} + \frac{1}{1 + nq_k} + \frac{1}{1 + \sqrt{d_1 d_2} np_j q_k} \right) \to 0. \tag{6}$$

Moreover, for $\nu \to \infty$, (6) *holds if and only if the following two conditions hold:*

$$\mathbb{E}\left[\frac{\#\{j \in [d_1] : R_j = 0\}}{d_1} \right] + \mathbb{E}\left[\frac{\#\{k \in [d_2] : C_k = 0\}}{d_2} \right] \to 0, \tag{7}$$

and for any deterministic index sets $S_n \subset [d_1] \times [d_2]$,

$$d_1 d_2 / |S_n| = O(1) \quad \Rightarrow \quad \mathbb{E}\big[\#\big\{ (j,k) \in S_n : X_{j,k} \geq 2 \big\} \big] \to \infty. \tag{8}$$

The parameters in Theorem 2.1 are all allowed to depend on the sample size n, $p_j = p_{j,n}$, $j \in [d_1]$, and $q_k = q_{k,n}$, $k \in [d_2]$, with $d_1 = d_{1,n}$ and $d_2 = d_{2,n}$. Thus, the theorem is uniform in $\{d_1, d_2, p_{1:d_1}, q_{1:d_2}\}$. For $\nu = (d_1 - 1)(d_2 - 1) = O(1)$, (6) is equivalent to the classic condition $\min_{j \in [d_1], k \in [d_2]} \mathbb{E}[X_{j,k}] \to \infty$. For $\nu \to \infty$, (7) and (8) give the statistical interpretation of the necessary and sufficient condition: The fractions of empty rows and columns converge to zero, and the expected number of cells with $X_{j,k} \geq 2$ diverges in any deterministic fractional subsets of cells.

Based on the analysis leading to the above theorem we develop adjusted degrees of freedom to improve the accuracy of the chi-squared approximation. This adjusted degrees of freedom is one half of the conditional variance of $\chi^2_{\text{cs-indep}}$ given the marginal totals $R_{1:d_1}$ and $C_{1:d_2}$ or the approximation of the conditional variance in the following proposition.

Proposition 2.2. *Let* $\chi^2_{\text{cs-indep}}$ *be as in* (4). *Then, under* (2), $\mathbb{E}[\chi^2_{\text{cs-indep}} | R_{1:d_1}, C_{1:d_2}] = 0$ *and the half conditional variance* $\overline{v}_{\text{cs}} = \text{Var}(\chi^2_{\text{cs-indep}} | R_{1:d_1}, C_{1:d_2})/2$ *can be expressed as*

$$\overline{v}_{\text{cs}} = \nu^* + \epsilon_n, \tag{9}$$

where

$$\nu^* = \frac{(d_1'' - 1)(d_2'' - 1) + (d_1' d_2' - 1)/n}{(1 - 1/n)(1 - 2/n)(1 - 3/n)} \tag{10}$$

with the d_1' *and* d_2' *in* (4), $d_1'' = \sum_{R_j > 0} (1 - 1/R_j)$ *and* $d_2'' = \sum_{C_j > 0} (1 - 1/C_j)$, *and*

$$\epsilon_n = \frac{(3\Delta_1\Delta_2 - n\Delta_1 - n\Delta_2) d_1' d_2' - n(d_1' + d_2' - 1)^2 - (d_1' - 1)^2 (d_2' - 1)^2 / (1 - 1/n)}{(n-1)(n-2)(n-3)},$$

with $\Delta_m = n(1 - d_m''/d_m') - d_m' \geq 0$, $m = 1, 2$. *Moreover,*

$$-\frac{2 - d_1''/d_1' - d_2''/d_2' - 1/n}{(n-1)(1-2/n)(1-3/n)} \leq \frac{\epsilon_n}{d_1' d_2'} \leq \frac{(1 - d_1''/d_1')(1 - d_2''/d_2')}{(n-1)(1-2/n)(1-3/n)}$$

for $n \geq d_1 \wedge d_2$, *and* $\epsilon_n \leq 0$ *when* $\Delta_1 + \Delta_2 \leq (4/3)n$.

Proposition 2.2 provides a formula for the conditional variance and a second order approximation of it with an explicit error bound. This second order approximation ν^* gives highly accurate results for the type I error rates in our simulation experiments as reported in Section 4.1. By definition,

$$\Delta_1 = \sum_{R_j > 0} \left(\frac{n}{R_j d_1'} + \frac{R_j d_1'}{n} - 2 \right), \quad \Delta_2 = \sum_{C_k > 0} \left(\frac{n}{C_k d_2'} + \frac{C_k d_2'}{n} - 2 \right). \tag{11}$$

We propose to use ν^* in (10) as adjusted degrees of freedom to the conventional degrees of freedom $\nu = (d_1 - 1)(d_2 - 1)$. The following theorem provides necessary and sufficient conditions for the validity of the degrees-of-freedom adjustment with ν replaced by ν^* or $\overline{\nu}_{\mathrm{CS}}$. Let $\chi^2_{\nu,1-\alpha}$ be the $(1-\alpha)$-th quantile of the χ^2_ν distribution, with $\chi^2_\nu \sim \mathrm{gamma}(\nu/2, 1/2)$ for all real $\nu > 0$.

Theorem 2.3. *Let ν^* and ϵ_n be as in Proposition 2.2. Suppose (2) and (7) hold. Then,*

$$\sup_{0\le\alpha\le 1}\left|\mathbb{P}\left\{\chi^2_{\text{CS-indep}} \ge \chi^2_{\widehat{\nu},1-\alpha} - \widehat{\nu}\right\} - \alpha\right| \to 0 \tag{12}$$

with either $\widehat{\nu} = \nu^*$ or $\widehat{\nu} = \overline{\nu}_{\mathrm{CS}}$ if and only if (5) holds with $\nu = (d_1 - 1)(d_2 - 1)$, if and only if

$$\frac{1}{d_1 d_2}\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\frac{1}{1+\sqrt{d_1 d_2}np_j q_k} \to 0. \tag{13}$$

Moreover, for $\nu \to \infty$, (13) is equivalent to (8).

With the degrees-of-freedom adjustment $\widehat{\nu}$, the $\chi^2_{\widehat{\nu}}$ approximation (12) matches the cross-sample chi-squared statistic in both the mean and the conditional variance, which clearly provides more accurate results in our simulation experiments as reported in Section 4.1. In this sense, the degrees-of-freedom adjusted version (12) is more robust than (5).

Because (7) is necessary for (5), the degrees-of-freedom adjusted chi-squared approximation always holds as in (12) when the classic chi-squared approximation holds as in (5). However, when (7) does not hold, the upper limit of $\mathbb{E}[\nu^*/\nu]$ is strictly less than 1, so that (5) must fail. Condition (7) holds if and only if

$$\frac{1}{d_1}\sum_{j=1}^{d_1}\frac{1}{1+np_j} + \frac{1}{d_2}\sum_{k=1}^{d_2}\frac{1}{1+nq_k} = o(1), \tag{14}$$

which implies $n \gg d_1 \vee d_2$ by Jensen's inequality. However, Theorem 2.1 allows $n \ll d_1 d_2$, so that zero may occupy a majority of the cells in the contingency table. Compared with (13), condition (8) is expressed in terms of the counts of the cells with $X_{j,k} \ge 2$. Thus, when the chi-squared approximation holds in a large two-way contingency table, one must expect to see a diverging number of cells with $X_{j,k} \ge 2$, but one may still see an overwhelming fraction of cells with $X_{j,k} = 0$.

Conditionally on marginal totals $R_{1:d_1} = r_{1:d_1}$ and $C_{1:d_2} = c_{1:d_2}$, the data matrix X has the hypergeometric distribution under the null hypothesis (2) with joint probability mass function

$$\overline{p}(x) = \frac{(r_1!\cdots r_{d_1}!)(c_1!\cdots c_{d_2}!)}{n!\prod_{j=1}^{d_1}\prod_{k=1}^{d_2}x_{j,k}!} \tag{15}$$

for the counting matrix $x = (x_{j,k})_{d_1\times d_2}$ with $r_j = \sum_{k=1}^{d_2} x_{j,k}$ and $c_k = \sum_{j=1}^{d_1} x_{j,k}$.

We are now ready to study the classic test statistic for independence in (3). This is done by comparing it with its cross-sample version in (4) through the following proposition.

Proposition 2.4. *Let χ^2_{indep} be Pearson's chi-squared statistic in (3) for the test of independence and $\chi^2_{\text{CS-indep}}$ be its cross-sample version in (4). Suppose (2) holds. Then,*

$$\mathbb{E}\left[\chi^2_{\text{indep}}\middle| R_{1:d_1}, C_{1:d_2}\right] = \frac{(d_1'-1)(d_2'-1)}{1-1/n} \tag{16}$$

with the d_1' and d_2' in (4), and the half conditional variance $\overline{\nu} = \mathrm{Var}(\chi^2_{\text{indep}}|R_{1:d_1}, C_{1:d_2})/2$ satisfies

$$\overline{\nu} = \overline{\nu}_{\mathrm{CS}} + \frac{1}{2}\left(1+\frac{4}{n-2}\right)\overline{V}_{\mathrm{diff}}, \tag{17}$$

where $\overline{V}_{\mathrm{diff}} = \mathrm{Var}(\chi^2_{\text{indep}} - \chi^2_{\text{CS-indep}}|R_{1:d_1}, C_{1:d_2}) = \Delta_1\Delta_2 d_1' d_2'/(n-1)$ with Δ_m $(m = 1, 2)$ in (11).

Proposition 2.4 gives an approximate orthogonal decomposition of Pearson's chi-squared statistic for the test of independence, as the difference $\chi^2_{\text{indep}} - \chi^2_{\text{cs-indep}}$ is nearly orthogonal to $\chi^2_{\text{cs-indep}}$. If (7) holds, $\text{Var}(\chi^2_{\text{cs-indep}}) = (1 + o(1))\nu$ by Proposition 2.2, so that $\overline{V}_{\text{diff}}$ is a small fraction of ν when

$$\text{Var}(\chi^2_{\text{indep}}) \le (2 + o(1))\nu. \tag{18}$$

This leads to the following theorem.

Theorem 2.5. *Suppose (2), (7) and (18) hold with $\nu = (d_1 - 1)(d_2 - 1)$. Then,*

$$\sup_{-\infty \le t \le \infty} \left| \mathbb{P}\left\{ \chi^2_{\text{indep}} - \frac{(d_1' - 1)(d_2' - 1)}{1 - 1/n} \le t \right\} - \mathbb{P}\left\{ \chi^2_\nu - \nu \le t \right\} \right| \to 0 \tag{19}$$

if and only if (13) holds, if and only if (8) holds, if and only if

$$\sup_{0 \le \alpha \le 1} \left| \mathbb{P}\left\{ \chi^2_{\text{indep}} - \frac{(d_1' - 1)(d_2' - 1)}{1 - 1/n} \ge \chi^2_{\widehat{\nu}, 1-\alpha} - \widehat{\nu} \right\} - \alpha \right| \to 0 \tag{20}$$

with $\widehat{\nu} = (d_1' - 1)(d_2' - 1)$ or $\widehat{\nu} = \overline{\nu}$, where $\overline{\nu}$ is the half conditional variance in (17).

It follows from Theorems 2.3 and 2.5 that (19), (20), (5) and (12) are all equivalent under (2), (7) and (18). However, the side condition (18), required only for the classic Pearson's chi-squared statistic, is noticeably more restrictive because approximately $\overline{V}_{\text{diff}}$ is expected to be

$$V_{\text{diff}} = \frac{d_1 d_2}{n} \sum_{j=1}^{d_1} \sum_{k=1}^{d_2} \left(\frac{1}{d_1 p_j} + d_1 p_j - 2 \right) \left(\frac{1}{d_2 q_k} + d_2 q_k - 2 \right)$$

under conditions (2) and (7). For example, when $p_j = (1 \pm \epsilon)/d_1$ and $q_k = (1 \pm \epsilon)/d_2$ with even d_1 and d_2 and $\epsilon = 1/\sqrt{2}$, $V_{\text{diff}} = (d_1 d_2)^2/n$, so that the requirement $V_{\text{diff}} = o(\nu)$ entails that $n \gg d_1 d_2$. From this point of view, the chi-squared approximation in (5) and (12) for the cross-sample statistic are more robust than their counterpart in (19) and (20) for the classic Pearson's chi-squared statistic.

Based on the results of this subsection, the approximation (12) is expected to be most accurate amongst the four methods. This is supported by our simulation results reported in Section 4.1. The proofs of Theorems 2.1, 2.3 and 2.5 are based on our study below of the one-way contingency table.

2.2 Power analysis for the cross sample version of chi-squared test of independence

We consider a probability $\mathbb{P}$ under which $X = (X_{j,k})_{d_1 \times d_2}$ ~multinomial$(n, (\theta_{j,k})_{d_1 \times d_2})$, and the null cell probabilities are $\kappa_{j,k} = p_j q_k$, $j \in [d_1]$, $k \in [d_2]$, where $p_j = \sum_{k=1}^{d_2} \theta_{j,k}$ and $q_k = \sum_{j=1}^{d_1} \theta_{j,k}$ are respectively the row and column marginal probabilities. We shall focus on the more stable cross-sample version of the test.

If the marginal probabilities p_j, q_k were known, the cross-sample chi-squared test statistic boils down to

$$\chi^2_{\text{CS}} = \sum_{j=1}^{d_1} \sum_{k=1}^{d_2} \frac{X^{(2)}_{j,k}}{n p_j q_k} - (n - 1). \tag{21}$$

Under $\mathbb{P}$, this χ^2_{CS} has mean

$$\mu_{\text{CS}} \quad = \quad (n-1) \sum_{j=1}^{d_1} \sum_{k=1}^{d_2} (\rho_{j,k} - 1)^2 p_j q_k - \sum_{j=1}^{d_1} (\overline{\rho}_{j,*} - 1)\big(\mathbb{P}\{R_j > 0\} - p_j\big)$$

$$-\sum_{k=1}^{d_w}(\overline{\rho}_{*,k}-1)\big(\mathbb{P}\{C_k>0\}-q_k\big)+\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2(\theta_{j,k}-p_jq_k) \tag{22}$$

with $\rho_{j,k}=\theta_{j,k}/(p_jq_k)$, $\overline{\rho}_{j,*}=\sum_{k=1}^{d_2}\rho_{j,k}\theta_{j,k}/p_j$ and $\overline{\rho}_{*,k}=\sum_{j=1}^{d_1}\rho_{j,k}\theta_{j,k}/q_k$, and variance

$$V_{\text{CS}}=\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\frac{\sigma_{\text{CS},j,k}^2}{n}+\big(1-1/n\big)\big(V_{\text{CS}}^*-V_{\text{row}}^*-V_{\text{col}}^*+V_{rc}^*\big) \tag{23}$$

with $\sigma_{\text{CS},j,k}=2(n-1)(\rho_{j,k}-\overline{\rho})\theta_{j,k}^{1/2}$, $\overline{\rho}=\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}\theta_{j,k}$,

$$\begin{aligned}
V_{\text{CS}}^*&=2\big(\overline{\rho}^2+\textstyle\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2(1-2\theta_{j,k})\big),\\
V_{\text{row}}^*&=4\textstyle\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2\theta_{j,k}(1/p_j-1)-2\sum_{j=1}^{d_1}\rho_{j,*}^2+2\overline{\rho}^2,\\
V_{\text{col}}^*&=4\textstyle\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2\theta_{j,k}(1/q_k-1)-2\sum_{k=1}^{d_2}\rho_{*,k}^2+2\overline{\rho}^2,
\end{aligned}$$

and $V_{rc}^*=V_{rc}'+4V_{rc}''-2V_{rc}'''+V_{rc}''''$ with

$$\begin{aligned}
V_{rc}'&=\textstyle\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}(\rho_{j,k}^2-\overline{\rho}_{*,k})(\rho_{j,k}^2-\overline{\rho}_{j,*})p_jq_k,\\
V_{rc}''&=\textstyle\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}(\rho_{j,k}-\overline{\rho}_{*,k})(\rho_{j,k}-\overline{\rho}_{j,*})(\rho_{j,k}-1)\theta_{j,k},\\
V_{rc}'''&=\textstyle\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\overline{\rho}_{j,*}\overline{\rho}_{*,k}(\rho_{j,k}-1)^2p_jq_k,\\
V_{rc}''''&=\textstyle\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\sum_{j'=1}^{d_1}\sum_{k'=1}^{d_2}\rho_{j,k}^2\rho_{j',k'}^2(\theta_{j,k'}-p_jq_{k'})(\theta_{j',k}-p_{j'}q_k).
\end{aligned}$$

If the null hypothesis is true, or equivalently $\theta_{j,k}=p_jq_k$ ($\rho_{j,k}=1$), for all j,k, we have $\mu_{\text{CS}}=0$ and $V_{\text{CS}}=2(1-1/n)(d_1d_2-1)$. When $\rho_{j,k}\approx 1$, the mean and variance of χ_{CS}^2 approximately match those of a non-central chi-squared distribution with $\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\sigma_{\text{CS},j,k}^2/n\approx 4\mu_{\text{CS}}$ as the contribution of the non-centrality to the variance. See section 2.4 for a power analysis of χ_{CS}^2.

In the case of most interest where the marginal probabilities are unknown, we approximate p_jq_k in (21) by R_jC_k/n^2 in the test statistic $\chi_{\text{CS-indep}}^2$ in (4). As a result, the test statistic is not a polynomial of the cell counts $X_{j,k}$, and the mean and variance of $\chi_{\text{CS-indep}}^2$ do not have simple expressions when $\theta_{j,k}\neq p_jq_k$. Thus, under (possibly alternative) probability $\mathbb{P}$, we approximate the mean of $\chi_{\text{CS-indep}}^2$ by

$$\mu_{\text{CS-indep}}=\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}(\rho_{j,k}-1)^2\Big((n+1)p_jq_k-q_k\mathbb{P}\{R_j>0\}-p_j\mathbb{P}\{C_k>0\}\Big), \tag{24}$$

and approximate the variance of $\chi_{\text{CS-indep}}^2$ by

$$\begin{aligned}
V_{\text{CS-indep}}\;=\;&\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\frac{\sigma_{j,k}^2}{n}+2\Big(1-\frac{1}{n}\Big)\Big(\overline{\rho}^2+\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2(1-2\theta_{j,k})\Big)\\
&-2\Big(1-\frac{1}{n}\Big)\Big(2\overline{\rho}^2+\sum_{j=1}^{d_1}\overline{\rho}_{j,*}^2(1-2p_j)+\sum_{k=1}^{d_2}\overline{\rho}_{*,k}^2(1-2q_k)\Big)\\
&+4\Big(1-\frac{1}{n}\Big)\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\overline{\rho}_{j,*}\overline{\rho}_{*,k}(\rho_{j,k}-1)^2p_jq_k,
\end{aligned} \tag{25}$$

where $\overline{\rho}_{j,*}=\sum_{k=1}^{d_2}\rho_{j,k}^2q_k$, $\overline{\rho}_{*,k}=\sum_{j=1}^{d_1}\rho_{j,k}^2p_j$, and $\sigma_{j,k}=(n-1)(2\rho_{j,k}-\overline{\rho}_{j,*}-\overline{\rho}_{*,k})\theta_{j,k}^{1/2}$. In fact, $\mu_{\text{CS-indep}}$ and $V_{\text{CS-indep}}$ are the mean and variance of a quadratic function of $X_{j,k}$ as an approximation of the

test statistic $\chi^2_{\text{CS-indep}}$. Both the approximate mean (24) and variance (25) exhibit row and column effects due to the use of estimated marginal probabilities in (21), leading to smaller values compared with (22) and (23) respectively. Under the null hypothesis of $\rho_{j,k} = 1, \forall\, j, k$, the above expressions reduce to $\mu_{\text{CS-indep}} = 0$ and $V_{\text{CS-indep}} = 2(1 - 1/n)(d_1 - 1)(d_2 - 1)$.

Consider the assumption below

$$\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\rho_{j,k}^2\theta_{j,k} + \sum_{k=1}^{d_2}\overline{\rho}_{*,k}^2 + \max_{1\le j\le d_1}\overline{\rho}_{j,*}^2 \ll V_{\text{CS-indep}}, \quad \sum_{j=1}^{d_1}\overline{\rho}_{j,*}^2 \lesssim V_{\text{CS-indep}}. \tag{26}$$

For general $\rho_{j,k}$, (26) implies

$$V_{\text{CS-indep}} \asymp V = \sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\left(\frac{\sigma_{j,k}^2}{n} + 2\rho_{j,k}^2\right). \tag{27}$$

Under the null hypothesis $\rho_{j,k} = 1, \forall j, k$, (26) is equivalent to $2 \le d_2$ and $d_1 \to \infty$.

In addition, similar to (14), we impose the following side conditions:

$$\begin{aligned}
&\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\left(\frac{\rho_{j,k}^2}{1 + n(p_j \wedge q_k)} + \left(\rho_{j,k}^2 - \overline{\rho}_{j,*} - \overline{\rho}_{*,k} + \overline{\rho}\right)^2 p_j q_k\right) \ll V_{\text{CS-indep}},\\
&\sum_{j=1}^{d_1}\sum_{k=1}^{d_2}\left((\rho_{j,k} - \overline{\rho}_{j,*})^2\frac{\theta_{j,k}}{p_j} + (\rho_{j,k} - \overline{\rho}_{*,k})^2\frac{\theta_{j,k}}{q_k}\right) \ll V_{\text{CS-indep}}, \quad \overline{\rho} = O(1).
\end{aligned} \tag{28}$$

Under the null hypothesis $\rho_{j,k} = 1, \forall j, k$, (28) is equivalent to (14).

Theorem 2.6. *Let $\chi^2_{\text{CS-indep}}$ be as in (4) with data $X = (X_{j,k})_{d_1\times d_2}$. Suppose $X \sim \text{multinomial}(n, (\theta_{j,k})_{d_1\times d_2})$ under a (possibly alternative) probability $\mathbb{P}$. Suppose (14), (26) and (28) hold and $d_1 d_2 \lesssim V$ for the quantity V in (27). Then,*

$$\sup_t \left|\mathbb{P}\left\{\chi^2_{\text{CS-indep}} - \mu_{\text{CS-indep}} \le t\right\} - \mathbb{P}\left\{\chi^2_{d'}(\tau') - (d' + \tau') \le t\right\}\right| \to 0 \tag{29}$$

for some (d', τ') satisfyiing $2d' + 4\tau' \ge (1 + o(1))V$ if and only if $(2d' + 4\tau')/V \to 1$ and

$$\max_{j,k}\frac{\Delta_{j,k}^2\lambda_{j,k} + \rho_{j,k}^2}{1 + \lambda_{j,k} \wedge (V/\rho_{j,k}^2)} + \sum_{j,k}\left(\frac{\Delta_{j,k}^4\lambda_{j,k}}{\Delta_{j,k}^2 + V} + \frac{\rho_{j,k}^4}{\rho_{j,k}^2 + \lambda_{j,k}^2 V}\right) \ll V, \tag{30}$$

where $\Delta_{j,k} = 2\rho_{j,k} - \overline{\rho}_{j,*} - \overline{\rho}_{*,k}$, $\lambda_{j,k} = n\theta_{j,k}$, V is as in (27), and the maximum and summation are taken over $(j, k) \in [d_1] \times [d_2]$.

2.3 Pearson's goodness-of-fit test

In a one-way contingency table with data $X_{1:d}$, Pearson's χ^2 statistic is by definition

$$\chi^2 = \sum_{j=1}^{d}\frac{(X_j - n\theta_j)^2}{n\theta_j} = \sum_{j=1}^{d}\frac{X_j^2}{n\theta_j} - n \tag{31}$$

with $\theta_j > 0$ and $\sum_{j=1}^{d}\theta_j = 1$. The cross-sample version of χ^2, recently studied in Xu et al. (2019), is

$$\chi^2_{\text{CS}} = \sum_{j=1}^{d}\frac{X_j(X_j - 1)}{n\theta_j} - n + 1. \tag{32}$$

Define quantities σ_j, $j \in [d]$, $\sigma > 0$ and V by

$$\sigma_j = \theta_j^{1/2}(1/\theta_j - d), \quad \sigma^2 = \sum_{j=1}^{d} \sigma_j^2, \quad V = \sigma^2/n + 2(d-1)(1-1/n). \tag{33}$$

As we will explain in the proof, $\sigma^2/n = \mathrm{Var}(\chi^2 - \chi^2_{\mathrm{CS}})$, and $\chi^2 - \chi^2_{\mathrm{CS}} = \sum_{j=1}^{d} X_j/(n\theta_j) - 1$, an average of i.i.d. random variables, is the linear part of Pearson's χ^2 statistic in its Hoeffding decomposition.

Theorem 2.7. *Let $(X_1, \ldots, X_d) \sim \mathrm{multinomial}(n, \theta_1, \ldots, \theta_d)$, where $d = d_n \geq 2$ and $\theta_j = \theta_{j,n}$ for $j \in [d]$ are allowed to depend on n. For $\alpha > 0$, let χ^2_α denote a $\mathrm{gamma}(\alpha/2, 1/2)$ random variable. Then, the mean and variance of Pearson's χ^2 in (31) are given by $\mathbb{E}[\chi^2] = d-1$ and $\mathrm{Var}(\chi^2) = V$. Moreover, with $d^* = V/2$ as adjusted degrees of freedom,*

$$\sup_t \left| \mathbb{P}\{\chi^2 - (d-1) \leq t\} - \mathbb{P}\{\chi^2_{d'} - d' \leq t\} \right| \to 0 \tag{34}$$

for some $d' \geq (1+o(1))d^*$ if and only if the following two conditions hold: $d'/d^* \to 1$ and

$$\frac{1}{d^*} \max_{1 \leq j \leq d} \frac{\sigma_j^2/n}{1+n\theta_j} + \frac{1}{d^*} \sum_{j=1}^{d} \left(\frac{\sigma_j^4/n^2}{\sigma_j^2/n + d^* n\theta_j} + \frac{1}{1+d^*(n\theta_j)^2} \right) \to 0. \tag{35}$$

The classic choice of d' is $d' = d-1$ to match the mean: $\mathbb{E}[\chi^2] = d-1$. This approximation is variance consistent if and only if $\sigma^2 \ll nd$. However, the condition $\sigma^2 \ll nd$ is nontrivial due to the singularity of σ^2 at small $n^{1/2}\theta_j$. Theorem 2.7 advocates the choice $d' = d^*$ to match the variance $\mathrm{Var}(\chi^2) = 2d^*$ after centering both Pearson's χ^2 and its approximation at the mean. This is a more natural approximation because $\chi^2 - (d-1)$ and $\chi^2_{d^*} - d^*$ match in both the mean and variance. For this reason, we call d^* adjusted degrees of freedom. Figure 1 in Section 3.3 demonstrates that $\mathrm{Var}(\chi^2)/2$ can be quite different from $\mathbb{E}[\chi^2]$. While (34) with $d' = d-1$ is commonly used, for general $d = d_n \to \infty$, the necessary and sufficient condition in Theorem 2.7 is new to the best of our knowledge.

As we will explain in the proof of Theorem 2.7, for fixed $d \geq 2$ or $2 \leq d = d_n = O(1)$, condition (35) is equivalent to $\min_{1 \leq j \leq d}(n\theta_j) \to \infty$ and implies $d^*/(d-1) \to 1$, so that its equivalence to (34) with $d' = d-1$ recovers the classical χ^2_{d-1} approximation theorem for Pearson's χ^2 statistic in (31).

Corollary 2.8. *For $\sigma^2 \ll nd$, the χ^2_{d-1} approximation*

$$\sup_t \left| \mathbb{P}\{\chi^2 \leq t\} - \mathbb{P}\{\chi^2_{d-1} \leq t\} \right| \to 0 \tag{36}$$

holds if and only if

$$\frac{1}{d} \sum_{j=1}^{d} \frac{1}{1+d^{1/2}n\theta_j} \to 0. \tag{37}$$

Corollary 2.8 can be seen as follows. Condition (36) implies $d^*/(d-1) = 1 - 1/n + \sigma^2/(n(d-1)) \to 1$ in view of Theorem 2.7. When $\sigma^2/(nd) \to 0$, (37) is equivalent to (35). Theorem 2.7 also provides, as a corollary, necessary and sufficient conditions for the normal approximation of Pearson's χ^2 statistic.

Corollary 2.9. *For $V' \geq (1+o(1))V$, the normal approximation*

$$\sup_t \left| \mathbb{P}\{\chi^2 - (d-1) \leq t\} - \mathbb{P}\{\mathcal{N}(0, V') \leq t\} \right| \to 0 \tag{38}$$

holds if and only if the following two conditions hold: $V'/V \to 1$ and

$$\frac{1}{d} + \frac{1}{d^*} \max_{1 \leq j \leq d} \frac{\sigma_j^2/n}{1+n\theta_j} + \frac{1}{d^*} \sum_{j=1}^{d} \left(\frac{\sigma_j^4/n^2}{\sigma_j^2/n + d^* n\theta_j} + \frac{1}{1+d^*(n\theta_j)^2} \right) \to 0. \tag{39}$$

Corollary 2.9 can be seen as follows. For $d = O(1)$, the normal approximation does not hold as $\chi^2 \geq 0$. For $d \to \infty$, $(\chi^2_{d^*} - d^*)/V^{1/2} \xrightarrow{\mathrm{D}} \mathcal{N}(0,1)$ as $V = 2d^* \to \infty$. Thus, compared with (34), the normal approximation just needs one more condition $d \to \infty$.

The χ^2_{d-1} and normal approximations were recently studied by Xu et al. (2019), who proved (36) under the sufficient condition

$$\frac{\sigma^2}{nd} + \frac{1}{d}\sum_{j=1}^{d} \frac{1}{(d^{1/2} n\theta_j)^{\delta}} \to 0 \tag{40}$$

for some $0 < \delta \leq 1$. For normal approximation, they also proved

$$\sup_t \left| \mathbb{P}\{\chi^2 - (d-1) \leq t\} - \mathbb{P}\{\mathcal{N}(0, \sigma^2/n) \leq t\} \right| \to 0 \tag{41}$$

under the sufficient condition

$$nd/\sigma^2 + \mathbb{E}\left[W_1^2 I\{W_1^2 \geq n^{1/2}\epsilon\}\right] \to 0 \tag{42}$$

with $W_1 = \sum_{j=1}^{d}(B_{1,j}/\theta_j - 1)/\sigma$ and $(B_{1,1}, \ldots, B_{1,d}) \sim \text{multinomial}(1, \theta_1, \ldots, \theta_d)$. While the second component of (42) is also necessary for (38) in the regime $nd \ll \sigma^2$, it is not necessary in the regime $\sigma^2 \ll nd$. As discussed in Xu et al. (2019), their approximation theory for Pearson's χ^2 statistics presents a dichotomy of the χ^2_{d-1} and $N(d-1, \sigma^2/n)$ regimes, which creates some extra difficulties in real applications. Their remedy is to use (32) as a diagonal-removed version (Bai and Saranadasa, 1996) of (31). Our analysis provides necessary and sufficient conditions for the centered χ^2_{d-1} approximation of (32).

Theorem 2.10. *Let χ^2_{CS} be as in (32) and $(X_1, \ldots, X_d) \sim \text{multinomial}(n, \theta_1, \ldots, \theta_d)$ with $d = d_n \geq 2$ and $\theta_j = \theta_{j,n}$ as in Theorem 2.7. Then, $\mathbb{E}\left[\chi^2_{\mathrm{CS}}\right] = 0$ and $\mathrm{Var}\left(\chi^2_{\mathrm{CS}}\right) = 2(d-1)(1-1/n)$. Moreover,*

$$\sup_t \left| \mathbb{P}\left\{\chi^2_{\mathrm{CS}} \leq t\right\} - \mathbb{P}\left\{\chi^2_{d-1} - (d-1) \leq t\right\} \right| \to 0 \tag{43}$$

if and only if

$$\frac{1}{d}\sum_{j=1}^{d} \frac{1}{1 + d(n\theta_j)^2} \to 0, \tag{44}$$

if and only if for all $S_n \subseteq \{1, \ldots, d\}$,

$$\frac{d}{|S_n|} = O(1) \quad \Rightarrow \quad \sum_{j \in S_n} \mathbb{P}\{X_j \geq 2\} \to \infty. \tag{45}$$

2.4 Power analysis for chi-squared goodness-of-fit tests

Suppose $X \sim \text{multinomial}(n, \theta_{1:d})$ and we want to test its goodness-of-fit against a null distribution with probability vector $\kappa_{1:d}$. Let

$$\mu = \sum_{j=1}^{d} \rho_j - \overline{\rho} + n(\overline{\rho} - 1) \tag{46}$$

with $\rho_j = \theta_j/\kappa_j$ and $\overline{\rho} = \sum_{j=1}^{d} \rho_j \theta_j$, and

$$V = \sum_{j=1}^{d} \frac{\sigma_j^2}{n} + 2\left(1 - \frac{1}{n}\right)\left(\overline{\rho}^2 + \sum_{j=1}^{d} \rho_j^2 (1 - 2\theta_j)\right) \tag{47}$$

with $\sigma_j = \{1/\kappa_j - \sum_{k=1}^d \rho_k + 2(n-1)(\rho_j - \overline{\rho})\}\theta_j^{1/2}$.

We call $\chi^2_\alpha(\tau)$ a non-central chi-square random variable if it has the density

$$\sum_{k=0}^{\infty} \frac{e^{-\tau/2}(\tau/2)^k}{k!} \frac{x^{\alpha/2+k-1}e^{-x/2}}{2^{\alpha/2+k}\mathrm{gamma}(\alpha+2k)}, \quad x > 0.$$

It is well understood that $\chi^2_\alpha(\tau)$ has mean $\alpha + \tau$ and variance $2\alpha + 4\tau$.

Theorem 2.11. *Let χ^2 be Pearson's chi-squared statistic in (31) with data $X_{1:d}$ and null parameter $\kappa_{1:d}$. Suppose $X_{1:d} \sim$ multinomial$(n, \theta_{1:d})$ under a probability $\mathbb{P}$ with (possibly alternative) parameter $\theta_{1:d}$. Then, $\mathbb{E}[\chi^2] = \mu$ with the μ in (46) and $\mathrm{Var}(\chi^2) = V$ with the V in (47). Moreover, when $\sum_{k=1}^d \rho_k^2\theta_k \ll V$,*

$$\sup_t \left| \mathbb{P}\{\chi^2 - \mu \le t\} - \mathbb{P}\{\chi^2_{d'}(\tau') - (d' + \tau') \le t\} \right| \to 0 \tag{48}$$

for some (d', τ') satisfying $2d' + 4\tau' \ge (1 + o(1))V$ if and only if $(2d' + 4\tau')/V \to 1$ and

$$\max_{1\le j\le d} \frac{\sigma_j^2/n + \rho_j^2}{1 + (n\theta_j) \wedge (V/\rho_j^2)} + \sum_{j=1}^d \left(\frac{\sigma_j^4/n^2}{\sigma_j^2/n + n\theta_j V} + \frac{\rho_j^4}{\rho_j^2 + (n\theta_j)^2 V} \right) \ll V. \tag{49}$$

where ρ_j and σ_j are as in (46) and (47) respectively.

When $\theta_{1:d} = \kappa_{1:d}$, $\rho_j = 1$ and $\sigma_j = (1/\kappa_j - d)\theta_j^{1/2}$, so that $\mu = d - 1$ and $V = 2(1 - 1/n)(d-1) + \sigma^2/n$ are as in Theorem 2.7 and the side condition $\sum_{k=1}^d \rho_k^2\theta_k \ll V$ becomes $1 \ll V$. Thus, Theorem 2.11 yields Theorem 2.7 in the case of $d = d_n \to \infty$.

For the cross-sample version of χ^2, let

$$\mu_{\mathrm{CS}} = (n-1)(\overline{\rho} - 1) = (n-1)\sum_{j=1}^d (\rho_j - 1)^2\kappa_j. \tag{50}$$

with $\rho_j = \theta_j/\kappa_j$ and $\overline{\rho} = \sum_{j=1}^d \rho_j\theta_j$ as in (46), and

$$V_{\mathrm{CS}} = \sum_{j=1}^d \frac{\sigma_{\mathrm{CS},j}^2}{n} + 2\left(1 - \frac{1}{n}\right)\left(\overline{\rho}^2 + \sum_{j=1}^d \rho_j^2(1 - 2\theta_j)\right). \tag{51}$$

with $\sigma_{\mathrm{CS},j} = 2(n-1)(\rho_j - \overline{\rho})\theta_j^{1/2}$.

Theorem 2.12. *Let χ^2_{CS} be the cross-sample chi-squared statistic in (32) with data $X_{1:d}$ and null parameter $\kappa_{1:d}$. Suppose $X_{1:d} \sim$ multinomial$(n, \theta_{1:d})$ under a probability $\mathbb{P}$ with (possibly alternative) parameter $\theta_{1:d}$. Then, $\mathbb{E}[\chi^2_{\mathrm{CS}}] = \mu_{\mathrm{CS}}$ with the μ_{CS} in (50), and $\mathrm{Var}(\chi^2_{\mathrm{CS}}) = V_{\mathrm{CS}}$ with the V_{CS} in (51). Moreover, under the condition $\sum_{k=1}^d \rho_k^2\theta_k \ll V_{\mathrm{CS}}$,*

$$\sup_t \left| \mathbb{P}\{\chi^2_{\mathrm{CS}} - \mu_{\mathrm{CS}} \le t\} - \mathbb{P}\{\chi^2_{d'}(\tau') - (d' + \tau') \le t\} \right| \to 0 \tag{52}$$

for some (d', τ') satisfyiing $2d' + 4\tau' \ge (1 + o(1))V_{\mathrm{CS}}$ if and only if $(2d' + 4\tau')/V_{\mathrm{CS}} \to 1$ and

$$\max_{1\le j\le d} \frac{n(\rho_j - \overline{\rho})^2\theta_j + \rho_j^2}{1 + (n\theta_j) \wedge (V_{\mathrm{CS}}/\rho_j^2)} + \sum_{j=1}^d \left(\frac{n(\rho_j - \overline{\rho})^4\theta_j}{(\rho_j - \overline{\rho})^2 + V_{\mathrm{CS}}} + \frac{\rho_j^4}{\rho_j^2 + (n\theta_j)^2 V_{\mathrm{CS}}} \right) \ll V_{\mathrm{CS}}. \tag{53}$$

Again, Theorem 2.12 yields Theorem 2.10 in the case of $d = d_n \to \infty$.

We write Pearson's chi-squared statistic as a second order U-statistic as follows. Let $B_i = B_{i,1:d}$ be independent multinomial$(1, \theta_{1:d})$ vectors and $X = X_{1:d} = \sum_{i=1}^n B_i$. Let $M = (M_{j,k})_{d\times d}$ with $M_{j,k} = I\{j = k\}/\theta_j - 1$. Assume $\min_{1\le j\le d} \theta_j > 0$. We have

$$\chi^2 = \sum_{j=1}^d \frac{X_j^2}{n\theta_j} - n = \frac{X^\top M X}{n} = \sum_{i_1=1}^n \sum_{i_2=1}^n \frac{B_{i_1}^\top M B_{i_2}}{n}$$

with $\mathbb{E}[\chi^2] = d - 1$. Hoeffding's decomposition of χ^2 is

$$\chi^2 = \chi^2_{CS} + \sum_{j=1}^d \left(\frac{X_j}{n\theta_j} - 1\right) + (d-1)$$

where χ^2_{CS}, the cross-sample version of χ^2, is the degenerate second order component,

$$\chi^2_{CS} = \chi^2 - \sum_j \frac{X_j}{n\theta_j} + 1 = \sum_{1\le i_1\ne i_2\le n} \frac{B_{i_1}^\top M B_{i_2}}{n}. \tag{54}$$

We apply Theorem **??** to approximate the distribution of χ^2_{CS}. We note that

$$M\theta_{1:d} = 0, \quad \mathbb{E}[MB_1B_1^\top M] = M, \quad \mathrm{Var}(B_1^\top M B_2) = \mathbb{E}[B_1^\top M B_1] = d - 1, \tag{55}$$

due to $M\mathbb{E}[B_1B_1^\top]M = M(\mathrm{diag}(\theta_{1:d}) - \theta_{1:d}\theta_{1:d}^\top)M = M - (\theta_{1:d}, \dots, \theta_{1:d})^\top M = M$. Let

$$\psi(x, y) = \frac{x^T M y}{\sqrt{d-1}}. \tag{56}$$

We find that $\mathrm{Var}(\psi(B_1, B_2)) = 1$ by (55), and in (**??**) with $\varepsilon_i = B_i$ and $v_n = 1 - 1/n$

$$W = \sum_{1\le i_1\ne i_2\le n} \frac{\sqrt{v_n}\psi(B_{i_1}, B_{i_2})}{\sqrt{2n(n-1)}} = \frac{\chi^2_{CS}}{\sqrt{2(d-1)}}. \tag{57}$$

The v_n here gives the right normalization as $2(d-1)$ is the variance of the target χ^2_{d-1} distribution. It also gives $\mathrm{Var}(\chi^2_{CS})/(2(d-1)) = \mathrm{Var}(W) = v_n$. By (**??**) and (55),

$$\overline{\psi}(x, y) = \frac{\psi(x, y)}{\sqrt{d-1}} = \frac{x^T M y}{d-1}, \tag{58}$$

so that $(d-1)^{-1/2}$ is the common nonzero eigenvalue of the operator ψ. The quantity $V_{T,\alpha}$ is nearly minimized with the shape parameter $\alpha = (d-1)/2$ of χ^2_{d-1}, giving

$$\begin{aligned} & \mathbb{E}\left[\left(T - 1 - W/\sqrt{\alpha}\right)^2\right] \\ \le\ & \mathrm{Var}\left(\sqrt{2}(2/n)\overline{\psi}(B_1, B_2)\right) + \mathrm{Var}\left(\overline{\psi}(B_1, B_1)\right)/n + (v_n - 1)^2 \\ =\ & \frac{8}{n^2(d-1)} + \frac{1}{n}\sum_{j=1}^d \frac{1/\theta_j - d}{(d-1)^2} + \frac{1}{n^2}. \end{aligned}$$

As $\theta_j > 0$ for all $j \le d$, this yields $\mathbb{E}\left[\left|T - 1 - W/\sqrt{\alpha}\right|\right] \lesssim 1/\sqrt{n}$ in (**??**) for fixed $\theta_{1:d}$. For general $\theta_{1:d}$ with diverging dimension, bounding separately the rare cell contribution would provide sharper upper bound for $\mathbb{E}\left[\left|T - 1 - W/\sqrt{\alpha}\right|\right]$. Moreover, specialized to (54), the terms $\overline{R}_1$ and $\overline{R}_2$ in Theorem **??** can be sharpened using the following lemma. This is an outline of the analysis leading to the L_1 bound for the χ^2_ν approximation of χ^2_{CS} in Theorem 2.14.

Lemma 2.13. *Let $X_{1:d} \sim$ multinomial$(n, \theta_{1:d})$ with $d \geq 2$ and $\theta_j > 0\, \forall j \in [d]$. Then,*

$$\begin{aligned}
\mathbb{E}\big[|X_j - n\theta_j|^3\big] &\leq 2n^{3/2}\big(\theta_j(1-\theta_j)\big)^{3/2} + n\theta_j(1-\theta_j), \quad \forall j, \\
\mathbb{E}\big[|X_j - n\theta_j|^2 |X_k - n\theta_k|\big] &\leq n^{3/2}\left(\theta_j\sqrt{\theta_k} + \frac{\theta_j^2(\theta_k - \sqrt{\theta_k})}{1-\theta_k}\right) + \frac{n\theta_j^2\theta_k}{1-\theta_k}, \quad j \neq k, \\
\sum_{j=1}^{d}\sum_{k=1}^{d} \mathbb{E}\big[(X_j - n\theta_j)^2 |X_k - n\theta_k|\big]/\theta_j &\leq (d\sqrt{d-1}+1)n^{3/2} + dn.
\end{aligned}$$

Theorem 2.14. *Let $d \geq 2$, χ^2_{CS} be as in (54), $\nu = d-1$, and χ^2_ν be a random variable with the chi-squared distribution with ν degrees of freedom. Then,*

$$d_{L_1}\left(\frac{\chi^2_{CS}}{\sqrt{2\nu}}, \frac{\chi^2_\nu - \nu}{\sqrt{2\nu}}\right) \leq C_0 \max\left\{\overline{R}_S^{(\nu+2)/(\nu+4)}, \overline{R}_S\right\} + \sqrt{V_{S,T}} + 4|S^c|/\nu. \tag{59}$$

where $C_0 = \sqrt{2/\pi} + 1$, $S = \{j : \sqrt{\nu} n\theta_j \geq c_0\}$ for some $c_0 > 0$ (e.g. $c_0 = 1$), and

$$\begin{aligned}
V_{S,T} &= \sum_{j\in S} \frac{1-\theta_j|S|}{\nu^2 n\theta_j} + \frac{1+8/\nu}{n^2}, \\
\overline{R}_S &= \frac{e\sqrt{8}}{\nu}\sum_{j\in S}\frac{\sqrt{n\theta_j}+1/2}{\sqrt{\nu}n\theta_j} + \frac{e\sqrt{8}}{n^{1/2}}\left(1 + \frac{1}{\nu} + \frac{1}{\nu^{3/2}} + \frac{1+1/\nu}{\sqrt{n\nu}}\right).
\end{aligned}$$

3 Log likelihood ratio and Hellinger statistic

The chi-squared and normal approximations are essentially equivalent for large contingency tables because the standardized chi-squared distribution is approximately Gaussian for large degrees-of-freedom. As classical Wilks' theorem has provided the chi-squared approximation of the log likelihood ratio when $d = O(1)$, we now focus on the normal approximation of the log likelihood ratio and Hellinger statistic when $d \to \infty$.

3.1 Log likelihood ratio goodness-of-fit test

Let $\lambda_j = n\theta_j$. By definition, the log likelihood ratio is given by

$$G = \sum_{j=1}^{d} f_{j,G}(X_j) = \sum_{j=1}^{d} 2X_j \log\big(X_j/\lambda_j\big), \tag{60}$$

where $f_{j,G}(x) = 2x\log(x/\lambda_j)$. Define

$$\sigma_{j,G} = (n\lambda_j)^{1/2}\sum_{k=1}^{d}\theta_k(\beta_j - \beta_k), \quad \sigma_G^2 = \sum_{j=1}^{d}\sigma_{j,G}^2, \quad V_G' = \sigma_G^2/n + \sum_{j=1}^{d} 1 \wedge \lambda_j^2, \tag{61}$$

where $\beta_j = \mathbb{E}[f_{j,G}(N_j+1) - f_{j,G}(N_j)]$ with $N_j \sim \mathrm{Pois}(\lambda_j)$ for $j \in [d]$. Similar to the σ^2 in (33), σ_G^2/n can be viewed as the variance of the linear part of the log likelihood ratio G. The following theorem is parallel to Corollary 2.9 for the asymptotic normality of G, i.e.,

$$\big(G - \mathbb{E}[G]\big)/\mathrm{Var}^{1/2}(G) \xrightarrow{\mathrm{D}} \mathcal{N}(0,1). \tag{62}$$

Theorem 3.1. *Let* $(X_1, \ldots, X_d) \sim \text{multinomial}(n, \theta_1, \ldots, \theta_d)$*, where* $d = d_n \geq 2$ *and* $\theta_j = \theta_{j,n}$ *for* $j \in [d]$ *are allowed to depend on n. Let* $V_G = \text{Var}(G)$ *be the variance of G and* V'_G *be as in* (61)*.*

(a) When $V_G \to \infty$*,* (62) *holds if and only if*

$$\frac{1}{d} + \frac{1}{V'_G} \max_{1 \leq j \leq d} \frac{\sigma^2_{j,G}/n}{1 + \lambda_j} + \frac{1}{V'_G} \sum_{j=1}^{d} \frac{\sigma^4_{j,G}/n^2}{\sigma^2_{j,G}/n + V'_G \lambda_j} \to 0. \tag{63}$$

(b) When $V_G \not\to \infty$ *and* $V_G \not\to 0$*,* (62) *holds if and only if*

$$\frac{1}{d} + \sum_{j=1}^{d} (1 - \theta_j)(1 \wedge \lambda_j^2) + \frac{1}{V'_G} \sum_{j=1}^{d} \frac{\sigma^4_{j,G}/n^2}{\sigma^2_{j,G}/n + V'_G \lambda_j} \to 0. \tag{64}$$

(c) When $V_G \to 0$ *and* $\limsup_{n\to\infty} \max_{1 \leq j \leq d} \theta_{j,n} < 1$*,* (62) *holds if and only if* $\sigma'^2_G > 0$ *and*

$$\frac{1}{d} + \sum_{j=1}^{d} \lambda_j^2 + \frac{1}{\sigma'^2_G} \sum_{j=1}^{d} \frac{\sigma'^4_{j,G}}{\sigma'^2_{j,G} + \sigma'^2_G \lambda_j} \to 0, \tag{65}$$

where $\sigma'_{j,G} = (n\lambda_j)^{1/2} \sum_{k=1}^{d} \theta_k \log(\theta_j/\theta_k)$ *and* $\sigma'^2_G = \sum_{j=1}^{d} \sigma'^2_{j,G}$*.*

Applying Steck's argument (Steck, 1955) for a central limit theory of conditional distributions, Morris (1975, Theorem 5.2) provided central limit theorems for Pearson's χ^2 and the log likelihood ratio allowing $d \to \infty$, under sufficient conditions that $\min_{1 \leq j \leq d} \lambda_j \geq \epsilon$ for some fixed $\epsilon > 0$ and $\max_{1 \leq j \leq d} \theta_j \to 0$. As we will see in Examples 3.4 and 3.5 in Section 3.3, neither of these two conditions are necessary.

3.2 Hellinger distance test for goodness of fit

Another closely related and important statistic is the Hellinger statistic

$$H = \sum_{j=1}^{d} f_{j,H}(X_j) = \sum_{j=1}^{d} 4(X_j^{1/2} - \lambda_j^{1/2})^2, \tag{66}$$

where $f_{j,H}(x) = 4(x^{1/2} - \lambda_j^{1/2})^2$. Similarly, define

$$\sigma_{j,H} = (n\lambda_j)^{1/2} \sum_{k=1}^{d} \theta_k (\gamma_j - \gamma_k), \quad \sigma^2_H = \sum_{j=1}^{d} \sigma^2_{j,H}, \quad V'_H = \sigma^2_H/n + \sum_{j=1}^{d} 1 \wedge \lambda_j^3, \tag{67}$$

where $\gamma_j = \mathbb{E}[f_{j,H}(N_j + 1) - f_{j,H}(N_j)]$ with $N_j \sim \text{Pois}(\lambda_j)$ for $j \in [d]$. Likewise, σ^2_H/n can be viewed as the variance of the linear part of the Hellinger statistic H. The following theorem provides necessary and sufficient conditions for the asymptotic normality of H, i.e.,

$$\big(H - \mathbb{E}[H]\big)/\text{Var}^{1/2}(H) \xrightarrow{\text{D}} \mathcal{N}(0, 1). \tag{68}$$

Theorem 3.2. *Let* $(X_1, \ldots, X_d) \sim \text{multinomial}(n, \theta_1, \ldots, \theta_d)$*, where* $d = d_n \geq 2$ *and* $\theta_j = \theta_{j,n}$ *for* $j \in [d]$ *are allowed to depend on n. Let* $V_H = \text{Var}(H)$ *be the variance of H and* V'_H *be as in* (67)*.*

(a) When $V_H \to \infty$*,* (68) *holds if and only if*

$$\frac{1}{d} + \frac{1}{V'_H} \max_{1 \leq j \leq d} \frac{\sigma^2_{j,H}/n}{1 + \lambda_j} + \frac{1}{V'_H} \sum_{j=1}^{d} \frac{\sigma^4_{j,H}/n^2}{\sigma^2_{j,H}/n + V'_H \lambda_j} \to 0. \tag{69}$$

(b) When $V_H \not\to \infty$ and $V_H \not\to 0$, (68) holds if and only if

$$\frac{1}{d} + \sum_{j=1}^{d} (1-\theta_j)(1 \wedge \lambda_j^4) + \frac{1}{V'_H} \sum_{j=1}^{d} \frac{\sigma_{j,H}^4/n^2}{\sigma_{j,H}^2/n + V'_H \lambda_j} \to 0. \tag{70}$$

(c) When $V_H \to 0$ and $\limsup_{n\to\infty} \max_{1\le j\le d} \theta_{j,n} < 1$, (68) holds if and only if

$$\frac{1}{d} + \sum_{j=1}^{d} \lambda_j^3 + \sum_{j=1}^{d} \frac{\lambda_j^3}{V''_H} \wedge \frac{\lambda_j^4}{V''^2_H} + \frac{1}{V''_H} \sum_{j=1}^{d} \frac{\sigma'^4_{j,H}/n^2}{\sigma'^2_{j,H}/n + V''_H \lambda_j} \to 0, \tag{71}$$

where $\sigma'_{j,H} = (n\lambda_j)^{1/2} \sum_{k=1}^{d} \theta_k(\eta_j - \eta_k)$ with $\eta_j = \lambda_j^{1/2}(2\lambda_j - 2 - \sqrt{2})$ for $j \in [d]$ and $V''_H = \sum_{j=1}^{d} \{\sigma'^2_{j,H}/n + \lambda_j^3\}$.

In Theorems 2.7, 3.1 and 3.2, the necessary and sufficient conditions involving σ_j^4, $\sigma_{j,G}^4$ or $\sigma_{j,H}^4$ are derived from Lindeberg's conditions for the linear and non-linear components of certain Poissonization of these statistics.

3.3 Examples

When $\theta_j = 1/d$ for $j \in [d]$, the multinomial model is called *equiprobable*.

Example 3.3 (Equiprobable)**.** *For equiprobable models, Pearson's χ^2, the log likelihood ratio and Hellinger statistics have approximately the χ^2_{d-1} distribution if and only if $n^2/d \to \infty$, and they are asymptotically normal if and only if $d \wedge (n^2/d) \to \infty$. In particular, for $n \ll d \ll n^2$, they are asymptotically normal even if $\lambda_j = n/d \to 0$.*

Pearson's χ^2, the log likelihood ratio and the Hellinger statistics can be asymptotically normal even when there is a large cell, e.g., $\theta_{1,n} \to 1$. Thus, the condition $\limsup_{n\to\infty} \max_{1\le j\le d} \theta_{j,n} < 1$ is *not* necessary in Theorem 3.1 (c) and Theorem 3.2 (c).

Example 3.4 (One large cell)**.** *Consider $(X_1, \dots, X_d) \sim \text{multinomial}(n, \theta_1, \dots, \theta_d)$ with*

$$\theta_1 = 1 - n^{-\alpha}, \quad \theta_2 = \cdots = \theta_d = n^{-\alpha}/(d-1),$$

where $\alpha \in (0,1)$ not depending on n and $d = d_n \to \infty$. In this case, for $d \ll n^2$, Pearson's χ^2 is asymptotically normal by Corollary 2.9. Moreover, for $d \gg n^{2(1-\alpha)}$, the log likelihood ratio G and the Hellinger statistic H are asymptotically normal by Theorem 3.1 (a) and Theorem 3.2 (a) respectively.

Moreover, these statistics can be asymptotically normal even when $n^2/d + \max_{1\le j\le d} \lambda_j \to 0$.

Example 3.5 (ϵ-Perturbation)**.** *Consider $(X_1, \dots, X_d) \sim \text{multinomial}(n, \theta_1, \dots, \theta_d)$ with*

$$\theta_1 = \cdots = \theta_{d/2} = (1+\epsilon)/d, \quad \theta_{d/2+1} = \cdots = \theta_d = (1-\epsilon)/d,$$

where $\epsilon = \epsilon_n \in (-1,1)\backslash\{0\}$ and $d = d_n \gg n^2$ is an even integer. In this case, for $\epsilon = 1/2$ in particular, Pearson's χ^2 statistic, the log likelihood ratio G and the Hellinger statistic H are asymptotically normal by Corollary 2.9, Theorem 3.1 (a) and Theorem 3.2 (a) respectively.

In Example 3.5, the log likelihood ratio G is asymptotically normal when $V'_G \to [0, \infty]$ in various cases. For instance, $V'_G \to \infty$ when ϵ is a constant not depending on n, $V'_G \asymp 1$ when $\epsilon = \tanh(n^{-1/2}/2)$, while $V'_G \to 0$ when $\epsilon = \tanh(n^{-1/2-\delta}/2)$ for a small constant $\delta > 0$ not depending on n.

For each statistic $T \in \{\chi^2, G, H\}$, we plot $\mathbb{E}[T]$ versus $\text{Var}(T)/2$ in the logarithmic scale for multinomial models in above examples. We set $d = \lceil n^{1.2} \rceil$ in all three examples, $\alpha = 1/2$ in Example 3.4, and $\epsilon = 1/2$ in Example 3.5. It can be seen from Figure 1 that $\text{Var}(T)/2$ is significantly different from $\mathbb{E}[T]$ in most cases. This supports the use of the degrees-of-freedom adjusted chi-squared approximation or the normal approximation to match both the mean and variance of these statistics.

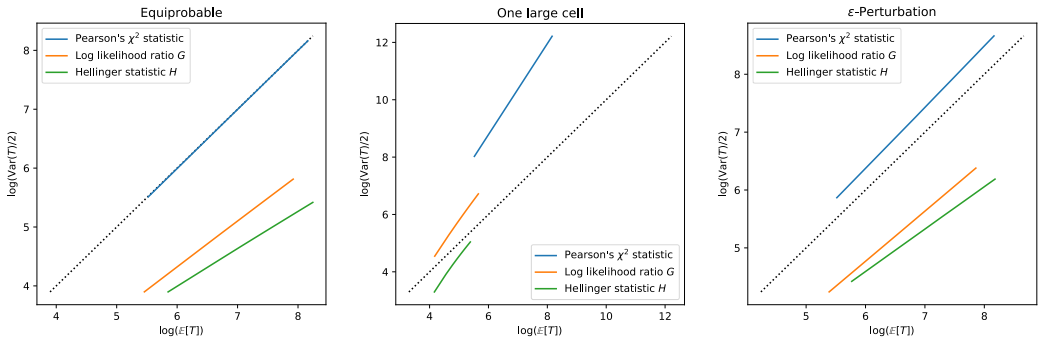


Figure 1: Mean versus half variance for χ^2, G and H in Examples 3.3, 3.4 and 3.5

4 Simulation results

4.1 Type I error rates comparison for tests of independence

We compare the performance in terms of the type I error rates of five approximation schemes, namely the chi-squared approximation for Pearson's χ^2_{indep} with degrees of freedom $\nu = (d_1-1)(d_2-1)$ and the adjusted degrees of freedom $\overline{\nu}$ in (17), the chi-squared approximation for the cross-sample chi-squared statistic $\chi^2_{\text{CS-indep}}$ with degrees of freedom $\nu = (d_1-1)(d_2-1)$ and the adjusted degrees of freedom ν^* and $\overline{\nu}_{\text{CS}}$ in (10) and (9).

In our simulations, we consider square contingency tables with $d_1 = d_2 = d$ and cell probabilities $p_j q_k$ generated in the following three cases:

- Case 1: For $j \in [d]$, sample i.i.d. $\widetilde{p}_j \sim \text{beta}(2,2)$ from a Beta distribution, and set the marginal row probability $p_j = \widetilde{p}_j / \sum_{j=1}^d \widetilde{p}_j$; Generate the marginal column probability $q_{1:d}$ the same way.
- Case 2: For $j \in [d]$, sample i.i.d. $\widetilde{p}_j \sim |\mathcal{N}(0,1)|$ from a folded normal distribution, and set the marginal row probability $p_j = \widetilde{p}_j / \sum_{j=1}^d \widetilde{p}_j$; Generate $q_{1:d}$ the same way.
- Case 3: Let $L = \lceil 0.7d \rceil$, sample i.i.d. $\widetilde{p}_j \sim \text{beta}(2,2)$ for $1 \le j \le L$, set $\widetilde{p}_j = 0.6^{j+1-L}$ for $L < j \le d$, and $p_j = \widetilde{p}_j / \sum_{j=1}^d \widetilde{p}_j$ for $j \in [d]$; Generate $q_{1:d}$ the same way.

Table 1: *Type I Error Rates, Case 1*

	df		χ^2_{indep}		$\chi^2_{\text{CS-indep}}$			χ^2_{indep}		$\chi^2_{\text{CS-indep}}$		
$(d, n/d)$	$\overline{\nu}$	$\overline{\nu}_{\text{CS}}$	χ^2_ν	$\chi^2_{\overline{\nu}}$	χ^2_ν	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$	χ^2_ν	$\chi^2_{\overline{\nu}}$	χ^2_ν	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$
	$\text{df} \times 10^{-4}$		$\alpha = 0.05$, $\hat{\alpha} \times 10^2$					$\alpha = 0.1$, $\hat{\alpha} \times 10$				
$(50, 5)$	0.440	0.109	0.98	6.11	1.52	**5.86**	5.88	0.15	1.05	0.39	**1.04**	1.04
$(70, 5)$	1.123	0.245	0.49	6.09	1.50	**5.91**	5.92	0.07	1.05	0.39	**1.05**	1.05
$(90, 5)$	2.289	0.35**6**	0.26	6.02	1.48	**5.95**	5.96	0.04	1.05	0.39	**1.05**	1.05
$(200, 10)$	16.60	2.73**5**	2.45	5.78	3.04	**5.63**	**5.63**	0.32	**0.99**	0.67	1.03	1.03
$(200, 50)$	6.241	3.714	8.00	**5.15**	4.63	5.17	5.17	1.28	0.97	0.94	**1.01**	**1.01**
$(400, 10)$	122.2	10.98	0.85	5.86	3.06	**5.63**	**5.63**	0.11	**1.02**	0.67	1.03	1.03
$(400, 50)$	35.24	14.93	9.85	5.17	4.62	**5.14**	**5.14**	1.44	0.97	0.93	**1.00**	**1.00**
$(600, 10)$	400.5	24.75	0.28	5.79	3.04	**5.63**	**5.63**	0.03	**1.02**	0.67	1.03	1.03
$(600, 50)$	102.3	33.64	11.0	5.25	4.63	**5.16**	**5.16**	1.52	0.97	0.94	**1.01**	**1.01**

Compared with Cases 1 and 2, the cell probabilities in Case 3 contain more very small elements, producing more sparse contingency tables.

In each of the three cases, we take 9 combinations of d and n and repeat the experiment independently for 10^6 times. In each experiment, we sample a random matrix $(X_{j,k})_{d\times d} \sim \text{multinomial}(n, (p_j q_k)_{d\times d})$, and compute the test statistics χ^2_{indep} and $\chi^2_{\text{CS-indep}}$. Based on the 10^6 realizations of these test statistics and for target test sizes $\alpha = 0.05$ and $\alpha = 0.1$, the proportions of false rejections are calculated and recorded in Tables 1, 2 and 3 for the three cases.

Table 2: *Type I Error Rates, Case 2*

	df		χ^2_{indep}		$\chi^2_{\text{CS-indep}}$			χ^2_{indep}		$\chi^2_{\text{CS-indep}}$		
$(d, n/d)$	$\overline{\nu}$	$\overline{\nu}_{\text{CS}}$	χ^2_{ν}	$\chi^2_{\overline{\nu}}$	χ^2_{ν}	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$	χ^2_{ν}	$\chi^2_{\overline{\nu}}$	χ^2_{ν}	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$
	$\text{df} \times 10^{-4}$		$\alpha = 0.05, \quad \hat{\alpha} \times 10^2$					$\alpha = 0.1, \quad \hat{\alpha} \times 10$				
$(50, 5)$	0.667	0.083	0.04	6.26	0.82	**5.81**	5.83	0.01	1.07	0.25	**1.03**	1.04
$(70, 5)$	1.767	0.16**4**	0.01	6.15	0.82	**5.88**	5.89	0.00	1.06	0.25	**1.03**	1.04
$(90, 5)$	3.678	0.27**3**	0.00	6.09	0.80	**5.88**	5.89	0.00	1.07	0.25	**1.04**	1.04
$(200, 10)$	47.90	2.183	0.03	6.23	2.02	**5.65**	**5.65**	0.00	1.05	0.50	**1.02**	**1.02**
$(200, 50)$	36.80	3.403	5.88	**5.11**	4.06	5.22	5.22	0.69	0.85	0.85	**1.01**	**1.01**
$(400, 10)$	375.1	8.76**9**	0.00	5.96	2.02	**5.70**	**5.70**	0.00	1.05	0.50	**1.03**	**1.03**
$(400, 50)$	281.8	13.68	3.83	5.34	4.14	**5.33**	**5.33**	0.43	0.89	0.86	**1.01**	**1.01**
$(600, 10)$	1257	19.76	0.00	5.84	2.03	**5.68**	**5.68**	0.00	1.04	0.49	**1.03**	**1.03**
$(600, 50)$	936.9	30.82	2.52	5.49	4.11	**5.25**	**5.25**	0.28	0.91	0.85	**1.01**	**1.01**

In each row in Tables 1, 2 and 3, the entry closest to the target value α is marked in bold. It can be observed that the degrees of freedom adjusted $\chi^2_{\nu^*}$ approximation of the cross-sample chi-squared statistic $\chi^2_{\text{CS-indep}}$ is the most robust scheme with the best overall size accuracy. For the classic χ^2_{indep} statistic, the degrees of freedom adjusted $\chi^2_{\overline{\nu}}$ approximation has much better size accuracy than the traditional χ^2_{ν} approximation.

Table 3: *Type I Error Rates, Case 3*

	df		χ^2_{indep}		$\chi^2_{\text{CS-indep}}$			χ^2_{indep}		$\chi^2_{\text{CS-indep}}$		
$(d, n/d)$	$\overline{\nu}$	$\overline{\nu}_{\text{CS}}$	χ^2_{ν}	$\chi^2_{\overline{\nu}}$	χ^2_{ν}	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$	χ^2_{ν}	$\chi^2_{\overline{\nu}}$	χ^2_{ν}	$\chi^2_{\nu^*}$	$\chi^2_{\overline{\nu}_{\text{CS}}}$
	$\text{df} \times 10^{-4}$		$\alpha = 0.05, \quad \hat{\alpha} \times 10^2$					$\alpha = 0.1, \quad \hat{\alpha} \times 10$				
$(50, 5)$	0.354	0.079	0.00	6.31	0.62	**5.67**	5.68	0.00	1.03	0.21	**1.03**	1.03
$(70, 5)$	0.746	0.151	0.00	6.29	0.54	**5.71**	5.72	0.00	1.04	0.19	**1.03**	1.03
$(90, 5)$	1.335	0.24**5**	0.00	6.29	0.49	**5.75**	5.75	0.00	1.04	0.18	**1.04**	1.04
$(200, 10)$	8.438	1.599	0.00	5.57	0.79	**5.48**	**5.48**	0.00	0.93	0.27	**1.03**	**1.03**
$(200, 50)$	10.23	2.045	0.00	3.53	1.27	**5.12**	**5.12**	0.00	0.61	0.39	**1.00**	**1.00**
$(400, 10)$	46.22	6.24**9**	0.00	5.66	0.71	**5.51**	**5.51**	0.00	0.97	0.26	**1.03**	**1.03**
$(400, 50)$	34.20	7.837	0.00	3.95	1.07	**5.14**	**5.14**	0.00	0.71	0.35	**1.01**	**1.01**
$(600, 10)$	133.9	13.95	0.00	5.71	0.69	**5.50**	**5.50**	0.00	**0.98**	0.25	1.03	1.03
$(600, 50)$	75.41	17.38	0.00	4.25	1.01	**5.16**	**5.16**	0.00	0.77	0.34	**1.01**	**1.01**

In our simulation experiments, ϵ_n is always negative and much smaller than ν^* in absolute value, so that the adjusted degrees of freedom ν^* in (9) provides a highly accurate approximation of the half conditional

variance $\overline{\nu}_{\text{CS}}$. In fact, their numerical values are virtually identical in the simulations except for a few cases, where the difference is only by one in the last digit marked in bold in the $\overline{\nu}_{\text{CS}}$-column of the tables.

As the degrees of freedom ν, $\overline{\nu}$ and $\overline{\nu}_{\text{CS}} \approx \nu^*$ are significantly different and diverge with d, the approximation schemes (12) and (20) matching both the conditional mean and variance of the test statistics are more robust and have better size accuracy throughout the low- and high-dimensional regimes as demonstrated by the numerical results. Since $\epsilon_n \leq 0$ in most cases and $\overline{\nu}_{\text{CS}} \leq \nu^*$, the $\chi^2_{\nu^*}$ approximation is slightly less conservative than the $\chi^2_{\overline{\nu}}$ approximation. Therefore, our simulations support the use of degrees of freedom adjusted $\chi^2_{\nu^*}$ approximation of the cross-sample $\chi^2_{\text{CS-indep}}$ statistic for the test of independence in large and sparse two-way contingency tables.

4.2 Power analysis for the tests of independence

To validate Theorem 2.6, we compare the empirical powers to the theoretical powers implied by Theorem 2.6. Two types of null cell probabilities $\{\kappa_{j,k} = p_j q_k\}_{j,k=1,\ldots,d}$, described below, are considered. The true cell probabilities $\{\theta_{j,k}\}_{j,k=1,\ldots,d}$ are drawn as follows.

- Case 1: Let $p'_j = q'_k = 1/d$ for all $j, k \in [d]$. Set for some $0 \leq a < 1$

$$\tilde{\theta}_{j,k} = \Big(1 + (-1)^{j+k-1} a\Big) p'_j q'_k.$$

  Finally, obtain $p_j = \sum_k \tilde{\theta}_{j,k}$, $q_k = \sum_j \tilde{\theta}_{j,k}$.

- Case 2: The $\{p'_j, q'_k\}_{j,k\in[d]}$ are generated in the same way as Case 1 in Section 4.1. Set for some $0 \leq a < 1$

$$\tilde{\theta}_{j,k} = \begin{cases} (1+a) p'_j q'_k & \text{if } j = k \\ (1-a) p'_j q'_k & \text{otherwise} \end{cases}$$

  Finally, normalize $\theta_{j,k} = \tilde{\theta}_{j,k} \big/ \big(\sum_{j,k} \tilde{\theta}_{j,k}\big)$ and obtain $p_j = \sum_k \theta_{j,k}$, $q_k = \sum_j \theta_{j,k}$.

The quantity $a > 1$ above measures the deviation of $\{\kappa_{j,k}\}$ from the true $\{\theta_{j,k}\}$. The larger a is, the larger is the difference between the null and alternative approximating distributions, and hence the higher is the power.

We vary (d, n, a) for each type of κ, and compare the empirical and theoretical powers in every combination $(\kappa_{1:d}, d, n, a)$. Given a $(\kappa_{1:d}, d, n, a)$, the empirical powers are computed by repeating the following steps independently for 1×10^4 times: sample the data vector $\{X_{j,k}\}$ from multinomial$(n, \theta_{j,k})$; compute the test statistic $\chi^2_{\text{CS-indep}}$; choose the cut-off values for $\chi^2_{\text{CS-indep}}$ at level α to be $\Phi^{-1}_{\nu^*}(1-\alpha) - \nu^*$ with $\Phi^{-1}_{\omega}(1-\alpha)$ being the $1-\alpha$ quantile of a chi-squared distribution with degrees of freedom ω; compare the test statistic $\chi^2_{\text{CS-indep}}$ with the cutoff value and either reject or accept the H_0. The distributions of the empirical powers at various a's are exhibited as black circles with error bars in Figure **??**. The theoretical powers are computed as follows. Theorem 2.6 states that when $\{\kappa_{j,k}\} \neq \{\theta_{j,k}\}$, the variance of $\chi^2_{\text{CS-indep}}$ matches the variance of a non-central chi-squared distribution $\chi^2_{\omega}(\tau)$ with degrees of freedom ω and noncentrality parameter τ, that is

$$V_{\text{CS-indep}} \approx 2\omega + 4\tau,$$

where $V_{\text{CS-indep}}$ is as in (27). We choose ω and τ to be

$$\omega = \min\{(d_1 - 1)(d_2 - 1), V_{\text{CS-indep}}/2\}, \quad \tau = V_{\text{CS-indep}}/4 - \omega/2.$$

The theoretical power of the cross-sample version test is

$$\begin{aligned}&\mathbb{P}\Big\{\chi^2_\omega(\tau) - (\omega+\tau) + \mu_{\text{CS-indep}} \geq \Phi^{-1}_{\nu^*}(1-\alpha) - \nu^*\Big\}\\ =&1 - \Phi_{\omega,\tau}\Big(\Phi^{-1}_{\nu^*}(1-\alpha) - \nu^* + (\omega+\tau) - \mu_{\text{CS-indep}}\Big),\end{aligned}$$

where $\mu_{\text{CS-indep}}$ is defined in (24), and $\Phi_{\omega,\tau}(\cdot)$ is the CDF of the non-central chi-square distribution $\chi^2_\omega(\tau)$.

Table 4: *Case 2, d=50, n=500*

theo_power	empi_power	theo_mean	emp_mean	theo_var_1	theo_var_2	emp_var
0.0868	0.0569	0.0000	-0.1743	4792.3960	4792.3960	3278.6538
0.0883	0.0544	0.5862	-0.2793	4800.6382	4802.527	3176.1545
0.0952	0.0617	3.1366	2.8063	4850.9603	4852.665	3353.1650
0.1160	0.0806	9.9226	9.5816	5010.1874	5004.261	3386.3866
0.1778	0.1327	26.6078	28.0681	5475.5768	5437.091	3656.8658
0.2140	0.1725	35.0971	36.6623	5739.0377	5679.849	3875.5176
0.2654	0.2203	46.3378	47.0818	6108.9494	6019.390	4059.1842
0.3378	0.3022	61.4026	64.0896	6636.3484	6501.882	4432.7396
0.4365	0.4199	81.9047	84.1355	7402.7595	7201.054	4842.0931
0.5629	0.5756	110.3493	113.5114	8543.0195	8238.858	5554.4964
0.7070	0.7673	150.7795	155.2109	10289.5114	9825.534	6394.6123
0.8431	0.9204	210.0305	217.7004	13063.0270	12341.958	8313.2097
0.9400	0.9891	300.3450	311.1163	17673.0000	16521.082	10866.8323
0.9862	0.9998	445.3187	461.5734	25798.7279	23883.356	15833.2849
0.9986	1.0000	695.0446	725.3056	41280.3646	37900.191	26479.4432

Table 5: *Case 1, d=50, n=1500*

theo_power	empi_power	theo_mean	emp_mean	theo_var	emp_var
0.0556	0.0496	0.0000	0.5065	4798.7987	4445.5975
0.0596	0.0549	2.2198	1.9191	4817.3315	4518.8960
0.0794	0.0743	11.9075	11.8614	4914.9108	4558.8649
0.1548	0.1485	37.7941	38.5428	5238.8198	4959.1122
0.4483	0.4501	101.8306	102.8487	6258.0830	5829.2130
0.5779	0.5731	128.2477	128.4753	6742.0577	6360.5395
0.7184	0.7218	161.5970	162.4110	7395.0197	6861.9399
0.8458	0.8587	204.0362	203.5686	8285.2528	7512.4663
0.9358	0.9529	258.5743	258.4397	9514.2919	8517.8534
0.9815	0.9879	329.4966	329.0423	11236.5718	10421.0512
0.9967	0.9985	423.0528	423.3693	13693.1374	12328.0230
0.9997	0.9999	548.6149	549.7539	17271.5813	15607.5997
1.0000	1.0000	720.7026	724.8403	22617.0429	20602.7122
1.0000	1.0000	962.7005	961.7051	30847.7117	27922.1457
1.0000	1.0000	1314.0816	1312.0225	43996.8987	39936.1803

We also compare the empirical powers of the five approximation schemes in Tables 1-3 and the U-statistic permutation test statistic (USP) proposed in (Berrett et al., 2021). The comparisons are made under four different cases. In each case, the empirical power is the average over 1×10^4 repetitions.

Table 6: *Case 1, d=150, n=4000*

theo_power	empi_power	theo_mean	emp_mean	theo_var	emp_var
0.0568	0.0486	0.0000	-1.8185	44390.8995	40259.7422
0.0579	0.0539	1.9944	4.1981	44424.6904	41490.0161
0.0632	0.0589	10.8044	11.4710	44589.9672	42183.1068
0.0796	0.0700	34.7646	34.2053	45102.6397	41156.0951
0.1345	0.1277	95.5743	97.3451	46636.0229	42684.6620
0.1635	0.1513	121.1346	120.9197	47351.4649	43429.8278
0.2052	0.1936	153.7311	153.8870	48312.7765	44404.7389
0.2661	0.2515	195.6918	193.4054	49621.4904	45520.1817
0.3544	0.3463	250.3283	252.1491	51431.3328	47474.2287
0.4782	0.4802	322.4641	325.2503	53981.5870	49016.7609
0.6371	0.6355	419.3199	418.7079	57656.8509	53574.5198
0.8067	0.8178	552.0591	553.7798	63098.9542	57807.1029
0.9353	0.9436	738.6123	738.7509	71427.8576	66386.4465
0.9901	0.9954	1009.1435	1010.3837	84705.8314	75911.8092
0.9996	1.0000	1417.3717	1416.5372	106989.9569	95678.9841

- Case 1: The cell probabilities are of the form

$$\theta_{j,k} = \frac{2^{-j-k}}{(1-2^{-d})^2} + a \cdot (\mathbb{1}_{j=k=1} + \mathbb{1}_{j=k=2}) - a \cdot (\mathbb{1}_{j=1,k=2} + \mathbb{1}_{j=2,k=1}),$$

  where $1 \le j, k \le d = 6$ and $a \ge 0$. The sample size is $n = 100$.

- Case 2: The cell probabilities are of the form

$$\theta_{j,k} = \frac{1}{d^2} + (-1)^{j+k-1} a,$$

  where $1 \le j, k \le d = 8$ and $a \ge 0$. The sample size is $n = 50$.

- Case 3: The $\{p'_j, q'_k\}_{j,k\in[d]}$ are generated in the same way as Case 1 in Section 4.1. Set for some $0 \le a < 1$

$$\tilde{\theta}_{j,k} = \begin{cases} (1+a)p'_j q'_k & \text{if } j = k \\ (1-a)p'_j q'_k & \text{otherwise} \end{cases}$$

  Finally, normalize $\theta_{j,k} = \tilde{\theta}_{j,k} \big/ \big(\sum_{j,k} \tilde{\theta}_{j,k}\big)$ and obtain $p_j = \sum_k \theta_{j,k}$, $q_k = \sum_j \theta_{j,k}$.

- Case 4: The $\{p'_j, q'_k\}_{j,k\in[d]}$ are generated in the same way as Case 3 in Section 4.1. Let $\mathcal{B} := \{(j,k) : p'_j q'_k \ge Q\}$, where Q is the 80% quantile of all $\{p'_j q'_k\}_{j,k}$. The cell probabilities are of the form

$$\tilde{\theta}_{j,k} = \Big(1 + (-1)^{j+k-1} a \cdot \mathbb{1}_{(j,k)\in\mathcal{B}}\Big) p'_j q'_k$$

  for some $0 \le a < 1$. Finally, normalize $\theta_{j,k} = \tilde{\theta}_{j,k} \big/ \big(\sum_{j,k} \tilde{\theta}_{j,k}\big)$.

Case 1 and 2 are the same as in Section 7.2 of Berrett et al. (2021). In Case 3, the departure from independence occurs in all cells, and is proportional to the cell probability. In Case 4, the alternative departs from independence in large-probability cells but not in small-probability cells.

As shown in Figure 2, the schemes "$\chi^2_{\text{CS-indep}}$ to approximate $\chi^2_{\nu^*}$" and "$\chi^2_{\text{CS-indep}}$ to approximate $\chi^2_{\bar{\nu}_{\text{CS}}}$" have comparable performances with the rate-optimal USP test in high-dimensional settings.

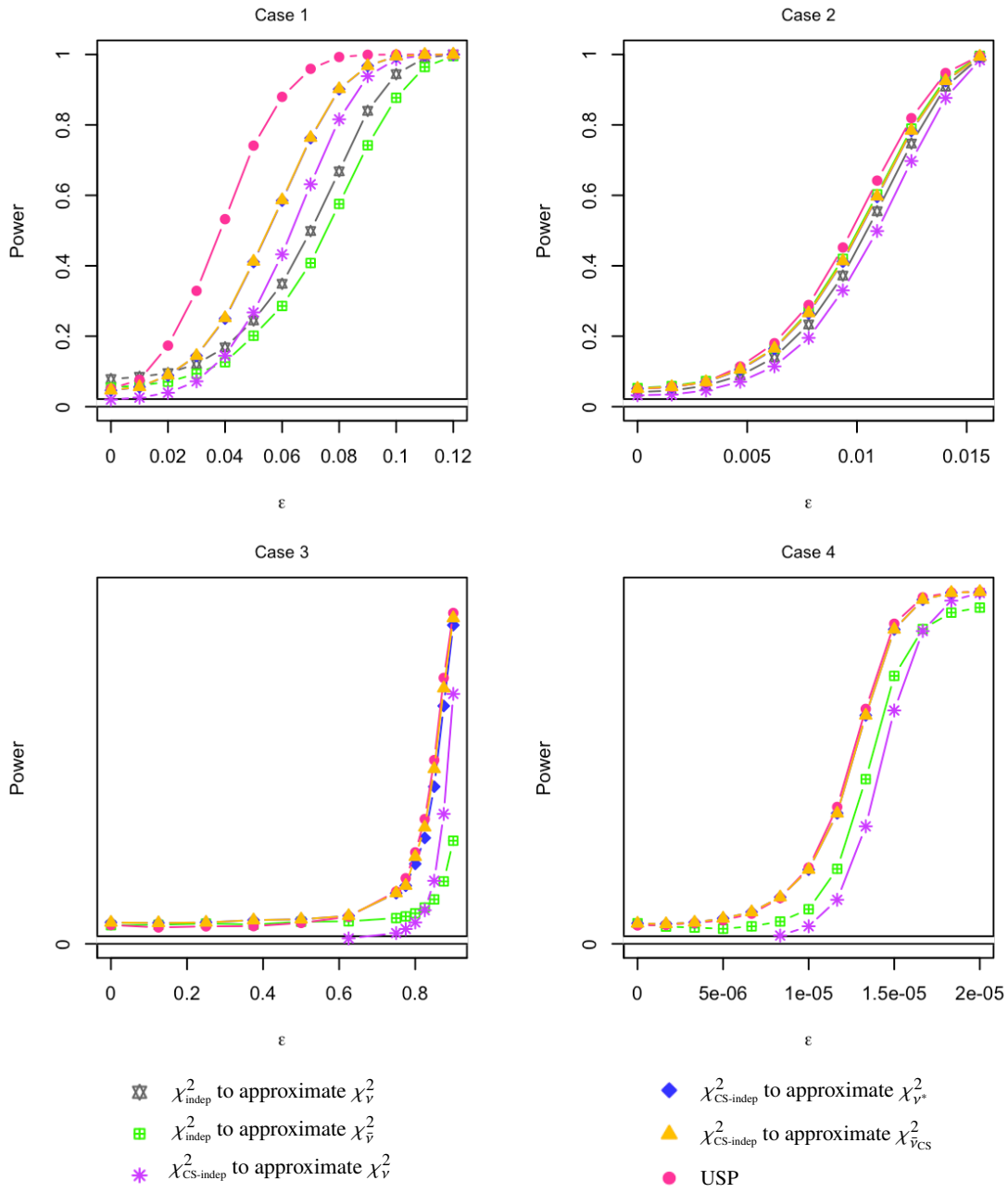


Figure 2: Comparison of the empirical powers of six tests. Each point represents average of 1×10^4 independent repetitions.

Table 7: *Case 2, d=50, n=1500*

theo_power	empi_power	theo_mean	emp_mean	theo_var	emp_var
0.0591	0.0511	0.0000	-0.6407	4798.7987	4283.1085
0.0630	0.0573	2.1266	2.4383	4815.2660	4347.3311
0.0827	0.0731	11.3814	12.0832	4907.9436	4433.8724
0.1554	0.1440	36.0126	35.8820	5216.5223	4668.7209
0.4325	0.4289	96.6033	98.2384	6179.8611	5534.9646
0.5562	0.5554	121.4997	122.5079	6634.2536	5990.5229
0.6934	0.6966	152.8636	153.3150	7245.0424	6462.5908
0.8232	0.8355	192.6833	191.1044	8074.3666	7065.0874
0.9207	0.9387	243.7220	244.6568	9214.1137	8187.1223
0.9747	0.9853	309.8986	309.7318	10803.2101	9509.6745
0.9948	0.9985	396.9038	397.9533	13057.0975	11414.7537
0.9994	0.9999	513.2290	513.1553	16319.7931	14447.3459
1.0000	1.0000	671.9567	671.7263	21159.6580	18434.6113
1.0000	1.0000	894.0259	896.5522	28554.1417	25308.1391
1.0000	1.0000	1214.5329	1215.7190	40266.7432	34882.2624

4.3 Type I error rates comparison for the goodness-of-fit tests

Under the same cases as in Xu et al. (2019), simulations are conducted to compare the size accuracy of the chi-squared and normal approximations of Pearson's χ^2, its cross-sample version χ^2_{CS}, the log likelihood ratio and Hellinger statistics. We consider two cases, in which the null distributions are respectively,

- Case 1 (Example 3.5 with $\epsilon = 1/2$): Set $\theta_j = 1/(2d)$ for $1 \le j \le d/2$, and set $\theta_j = 3/(2d)$ for $d/2+1 \le j \le d$.
- Case 2: Set $\theta_j = 2^{-d/2}(d/2-1)!\big/\big((j-1)!(d/2-j)!\big)$ for $1 \le j \le d/2$, and set $\theta_j = 1/d$ for $d/2+1 \le j \le d$.

In each case, we let (n, d) vary in $\{40, 100, 400\}$ and repeat the experiment independently for 6×10^6 times. In each experiment, we sample $(X_1, \ldots, X_d) \sim \text{multinomial}(n, \theta_1, \ldots, \theta_d)$ and compute the test statistics χ^2, χ^2_{CS}, G and H under the null hypothesis. For target significance level $\alpha = 0.05$, type I error rates are calculated and recorded in Tables 9 and 10.

We compare the performances of the normal approximation, the chi-squared approximation with degrees of freedom matching the mean, and the centered chi-squared approximation with degrees of freedom matching the half variance. For simplicity, the two chi-squared approximation schemes are called "traditional" and "adjusted" respectively. For Pearson's χ^2, the log likelihood ratio G and the Hellinger statistic H, all three approximation schemes are considered. For the cross sample chi-squared statistic χ^2_{CS}, the half variance is $(d-1)(1-1/n)$, so that only the normal and χ^2_{d-1} approximations are considered. This gives totally 11 approximation schemes for the 4 test statistics. The three best results are marked in bold in each row.

In Table 9, where the cell probability θ_j's are relatively balanced in Case 1, the adjusted chi-squared approximation is the best overall performer, followed by the normal approximation. This is expected because the two better methods match test statistics in both the mean and variance. Among the four statistics, Pearson's χ^2 is over all the least stable and demonstrate a fatter right tail than the chi-squared distribution. This is probably caused by the singularities at small cell probability θ_j's in the denominators.

In Table 10, where the cell probabilities are unbalanced with some very small θ_j's in Case 2, the log likelihood ratio demonstrates most stable results overall, followed by the Hellinger statistic. Again, the adjusted chi-squared approximation performs the best, followed by the normal approximation.

Table 8: *Case 2, d=150, n=4000*

theo_power	empi_power	theo_mean	emp_mean	theo_var	emp_var
0.0608	0.0530	-0.0000	2.9718	44390.8995	39333.2587
0.0620	0.0568	1.9543	0.4743	44421.9310	39577.4328
0.0674	0.0605	10.5792	14.4407	44582.9187	39552.2174
0.0842	0.0720	34.0034	30.8264	45086.5525	40832.0124
0.1395	0.1219	93.3331	93.3404	46592.9960	40334.2526
0.1683	0.1580	118.2347	123.0909	47294.6934	41775.8814
0.2096	0.1976	149.9659	152.0386	48236.4061	43459.3135
0.2693	0.2497	190.7759	188.5374	49516.5749	44484.2637
0.3555	0.3389	243.8593	242.0095	51283.8702	44706.4035
0.4759	0.4718	313.8613	315.8478	53769.0492	47755.3680
0.6307	0.6348	407.7220	409.7099	57341.8061	50041.9130
0.7980	0.8107	536.1459	535.9399	62616.7991	56240.0845
0.9289	0.9400	716.2806	713.1279	70661.9108	62392.7853
0.9881	0.9947	976.8780	979.6849	83433.3907	72325.4295
0.9994	0.9999	1368.9406	1367.7441	104755.8693	92907.9815

Table 9: *Type I Error Rates* $\times 10^2$ *at* $\alpha = 0.05$*, Case 1*

	Pearson's χ^2			Cross-sample χ^2_{CS}		Log likelihood ratio G			Hellinger H		
(d, n)	$\mathcal{N}(0,1)$	χ^2_{d-1}	$\chi^2_{V/2}$	$\mathcal{N}(0,1)$	χ^2_{d-1}	$\mathcal{N}(0,1)$	$\chi^2_{\mu_G}$	$\chi^2_{V_G/2}$	$\mathcal{N}(0,1)$	$\chi^2_{\mu_H}$	$\chi^2_{V_H/2}$
(40, 40)	6.86	6.86	5.64	5.50	**5.50**	5.67	1.80	**4.51**	5.50	3.46	**4.58**
(40, 100)	6.30	5.77	5.29	6.24	**5.18**	5.91	4.17	**4.83**	5.99	8.97	**5.27**
(40, 400)	6.08	5.14	**5.02**	6.05	**4.98**	6.10	5.41	**5.06**	6.69	8.87	5.88
(100, 40)	7.04	9.27	**5.31**	6.30	6.30	**5.35**	0.24	4.16	**5.24**	0.06	4.07
(100, 100)	6.43	6.43	5.62	5.95	**5.10**	5.55	1.78	4.65	**5.34**	3.49	**4.70**
(100, 400)	5.90	5.46	**5.04**	5.74	**5.08**	5.61	**5.05**	4.90	5.69	10.60	5.21
(400, 40)	6.28	14.40	6.28	5.96	5.96	**5.63**	0.00	**4.30**	**5.34**	0.00	3.47
(400, 100)	6.39	9.99	6.39	5.57	**5.57**	**5.31**	0.04	4.40	**5.18**	0.00	4.20
(400, 400)	5.83	6.77	5.40	5.38	5.38	5.29	1.76	**4.82**	**5.17**	3.56	**4.83**

4.4 Power analysis for the tests of goodness-of-fit

We conduct some numerical experiments to verify the accuracy of the approximate power functions implied by Theorems 2.11 and 2.12. Two types of $\kappa_{1:d}$'s (null probability vectors) are considered and described below.

- Case 1: Set $\bar{\kappa}_j = 1/d + (1/6d) \cdot z_j$ with $z_j \overset{iid}{\sim} Bernoulli(1/2)$ for all $1 \le j \le d$, and normalize to get $\kappa_j = \bar{\kappa}_j/(\sum_j \bar{\kappa}_j)$. Set $\bar{\theta}_j = \kappa_j/a$ for $1 \le j \le d/2$, $\bar{\theta}_j = (2 - 1/a)\kappa_j$ for $d/2 + 1 \le j \le d$ with some $a > 1$, and normalize to get $\theta_j = \bar{\theta}_j/(\sum_j \bar{\theta}_j)$.

- Case 2: Set $\kappa_j = 2^{-d/2} \frac{(d/2-1)!}{(j-1)!(d/2-j)!}$ for $1 \le j \le d/2$, and $\kappa_j = 1/d$ for $d/2 + 1 \le j \le d$. Set $\theta_j = (2 - 1/a)\kappa_j$ for $1 \le j \le d/2$, and $\theta_j = \kappa_j/a$ for $d/2 + 1 \le j \le d$ with $a > 1$.

The positive number a above quantifies the deviation of $\kappa_{1:d}$ from the true $\theta_{1:d}$. It can be shown that $\bar{\rho}$ and μ in (46), and μ_{CS} in (50) are increasing in a when $a > 1$.

Table 10: *Type I Error Rates* $\times 10^2$ *at* $\alpha = 0.05$*, Case 2*

	Pearson's χ^2			Cross-sample χ^2_{CS}		Log likelihood ratio G			Hellinger H		
(d, n)	$\mathcal{N}(0,1)$	χ^2_{d-1}	$\chi^2_{V/2}$	$\mathcal{N}(0,1)$	χ^2_{d-1}	$\mathcal{N}(0,1)$	$\chi^2_{\mu_G}$	$\chi^2_{V_G/2}$	$\mathcal{N}(0,1)$	$\chi^2_{\mu_H}$	$\chi^2_{V_H/2}$
(40, 40)	0.16	6.48	0.16	**4.67**	3.73	5.88	2.55	**4.58**	5.64	4.28	**4.67**
(40, 100)	0.38	6.08	0.38	**4.92**	4.00	6.00	**4.99**	**4.88**	6.09	10.65	5.32
(40, 400)	1.24	5.38	1.18	**5.18**	4.27	6.16	**4.98**	**5.04**	6.16	6.16	5.18
(100, 40)	0.00	3.86	0.00	3.92	3.31	6.04	0.87	**4.80**	**5.93**	0.77	**4.84**
(100, 100)	0.00	3.61	0.00	3.79	3.23	5.64	2.40	**4.68**	**5.43**	4.07	**4.73**
(100, 400)	0.00	3.57	0.00	3.78	3.24	5.82	**5.43**	**5.03**	6.21	10.90	**5.65**
(400, 40)	0.00	1.97	2.66	3.57	3.23	**5.52**	0.60	**4.54**	**5.34**	0.15	4.33
(400, 100)	0.00	1.95	2.50	2.92	2.65	**5.77**	0.60	**4.94**	5.89	0.48	**5.13**
(400, 400)	0.00	1.69	2.49	2.58	2.34	5.33	2.09	**4.77**	**5.16**	3.71	**4.76**

We vary (d, n, a) for each type of κ, and compare the empirical and theoretical powers in every setting. Given a $(\kappa_{1:d}, d, n, a)$, the empirical powers are computed by repeating the following steps independently for 1×10^4 times: sample the data vector $(X_1, \ldots, X_d)$ from multinomial$(n, \theta_1, \ldots, \theta_d)$; compute the test statistics χ^2, χ^2_{CS} under the null probability vector $\kappa_{1:d}$; choose the cut-off values for χ^2 $-(d-1)$ and χ^2_{CS} at level α to be, respectively, $\Phi^{-1}_{V/2}(1-\alpha) - V/2$ and $\Phi^{-1}_{d-1}(1-\alpha) - (d-1)$ with $\Phi^{-1}_{\omega}(1-\alpha)$ being the $1-\alpha$ quantile of χ^2_ω distribution and V as in (33). The theoretical powers are computed as follows. Theorem 2.11 states that when $\kappa_{1:d} \neq \theta_{1:d}$, the variance of χ^2 approximates the variance of a non-central chi-squared distribution $\chi^2_\omega(\tau)$ with degrees of freedom ω and noncentrality parameter τ, that is

$$2\omega + 4\tau \approx V',$$

where V' is as in (47). We choose ω and τ as

$$\omega = V'/2, \quad \tau = 0.$$

The theoretical power of the Pearson's chi-square test is

$$\begin{aligned}
&\mathbb{P}\Big\{\chi^2_\omega(\tau) - (\omega+\tau) + \mu - (d-1) \geq \Phi^{-1}_{V/2}(1-\alpha) - V/2\Big\} \\
=&1 - \Phi_{\omega,\tau}\Big(\Phi^{-1}_{V/2}(1-\alpha) - V/2 + (d-1) - \mu + (\omega+\tau)\Big) \\
=&1 - \Phi_{\omega,\tau}\Big(\Phi^{-1}_{V/2}(1-\alpha) - V/2 + 3(d-1)/2 + V'/4 - \mu\Big),
\end{aligned}$$

where $\Phi_{\omega,\tau}(\cdot)$ is the CDF of the non-central chi-square distribution $\chi^2_\omega(\tau)$, and μ is as in (46). Similarly, by Theorem 2.12, the variance of χ^2_{CS} approximates the variance of a non-central chi-squared distribution $\chi^2_{\omega_{\text{CS}}}(\tau_{\text{CS}})$ with degrees of freedom ω_{CS} and noncentrality parameter τ_{CS} satisfying

$$2\omega_{\text{CS}} + 4\tau_{\text{CS}} = V_{\text{CS}},$$

where V_{CS} is as in (51). Picking ω_{CS} and τ_{CS} as

$$\omega_{\text{CS}} = V_{\text{CS}}/2, \quad \tau_{\text{CS}} = 0$$

the theoretical power of the cross-sample version test is

$$\begin{aligned}
&\mathbb{P}\left\{\chi^2_{\omega_{\mathrm{CS}}}(\tau_{\mathrm{CS}}) - (\omega_{\mathrm{CS}} + \tau_{\mathrm{CS}}) + \mu_{\mathrm{CS}} \geq \Phi^{-1}_{d-1}(1-\alpha) - (d-1)\right\} \\
=&1 - \Phi_{\omega_{\mathrm{CS}},\tau_{\mathrm{CS}}}\left(\Phi^{-1}_{d-1}(1-\alpha) - (d-1) + (\omega_{\mathrm{CS}} + \tau_{\mathrm{CS}}) - \mu_{\mathrm{CS}}\right) \\
=&1 - \Phi_{\omega_{\mathrm{CS}},\tau_{\mathrm{CS}}}\left(\Phi^{-1}_{d-1}(1-\alpha) + V_{\mathrm{CS}}/4 - (d-1)/2 - \mu_{\mathrm{CS}}\right)
\end{aligned}$$

with μ_{CS} as in (50). When $\theta_{1:d} = \kappa_{1:d}$, the theoretical powers for both tests are α.

As shown in Figure 3 and Figure 4, the cross-sample version statistic is more stable and robust than Pearson's chi-squared statistic. For the cross-sample version, the theoretical powers (orange) predict accurately the empirical powers (black) across all settings. While Pearson's chi-squared statistic fails when the cell probabilities are unbalanced.

5 Discussion

This paper develops a sharp distributional theory for Pearson's chi-squared statistic when the number of cells and the cell probabilities vary with the sample size. Under the assumptions of the respective theorems, the necessary and sufficient conditions characterize the validity of chi-squared and normal approximations across classical, sparse, and unbalanced multinomial regimes. Necessity makes the theory informative about both validity and failure: it identifies the features of a probability array that obstruct an approximation and clarifies which familiar sufficient conditions can be relaxed. The resulting perspective places distributional approximation at the center of the analysis, with goodness-of-fit and independence testing providing applications.

One implication is that a large fraction of empty cells can coexist with an accurate approximation. For the cross-sample statistic with diverging dimension, the occupancy criterion requires the expected number of cells containing at least two observations to diverge in every deterministic subset occupying a nonvanishing fraction of the table. This condition describes how repeated occupancy supports the quadratic fluctuations of the statistic even when most cells are empty. In the equiprobable case, the boundary $n^2/d \to \infty$ permits the regime $n \ll d \ll n^2$, where every expected count vanishes. For nonuniform probabilities, additional fluctuations can produce asymptotic normality even beyond this dimension range. Thus, both the allocation of probability mass and the source of the variance matter when assessing sparsity.

Probability imbalance also separates the mean from the variance of Pearson's statistic. Its exact mean remains $d-1$, while its variance can differ substantially from $2(d-1)$. The adjusted degrees of freedom $d^\star = \mathrm{Var}(\chi^2)/2$ accommodate this discrepancy through the centered approximation of $\chi^2 - (d-1)$ by $\chi^2_{d^\star} - d^\star$. This construction preserves the contribution of unequal cell probabilities and connects chi-squared and normal approximations as the adjusted degrees of freedom diverge. Moment matching alone, however, does not guarantee distributional accuracy; the conditions in the approximation theorems remain essential. The simulation results illustrate the practical value of distinguishing variance mismatch from other causes of approximation failure.

The decompositions used in the proofs help explain these different behaviors. Separating linear and quadratic components reveals how probability imbalance and repeated occupancy contribute to the fluctuations, while Poissonization makes the relevant Lindeberg-type conditions explicit. Removing the linear component gives the cross-sample statistic a simpler variance structure and an interpretable validity criterion. The corresponding results for the log likelihood ratio and Hellinger statistics extend this analysis to other multinomial functionals under the stated variance regimes and side conditions. The two-way-table results show how the framework accommodates estimated marginal probabilities, and the analysis under alternatives connects the distributional approximations to power calculations.

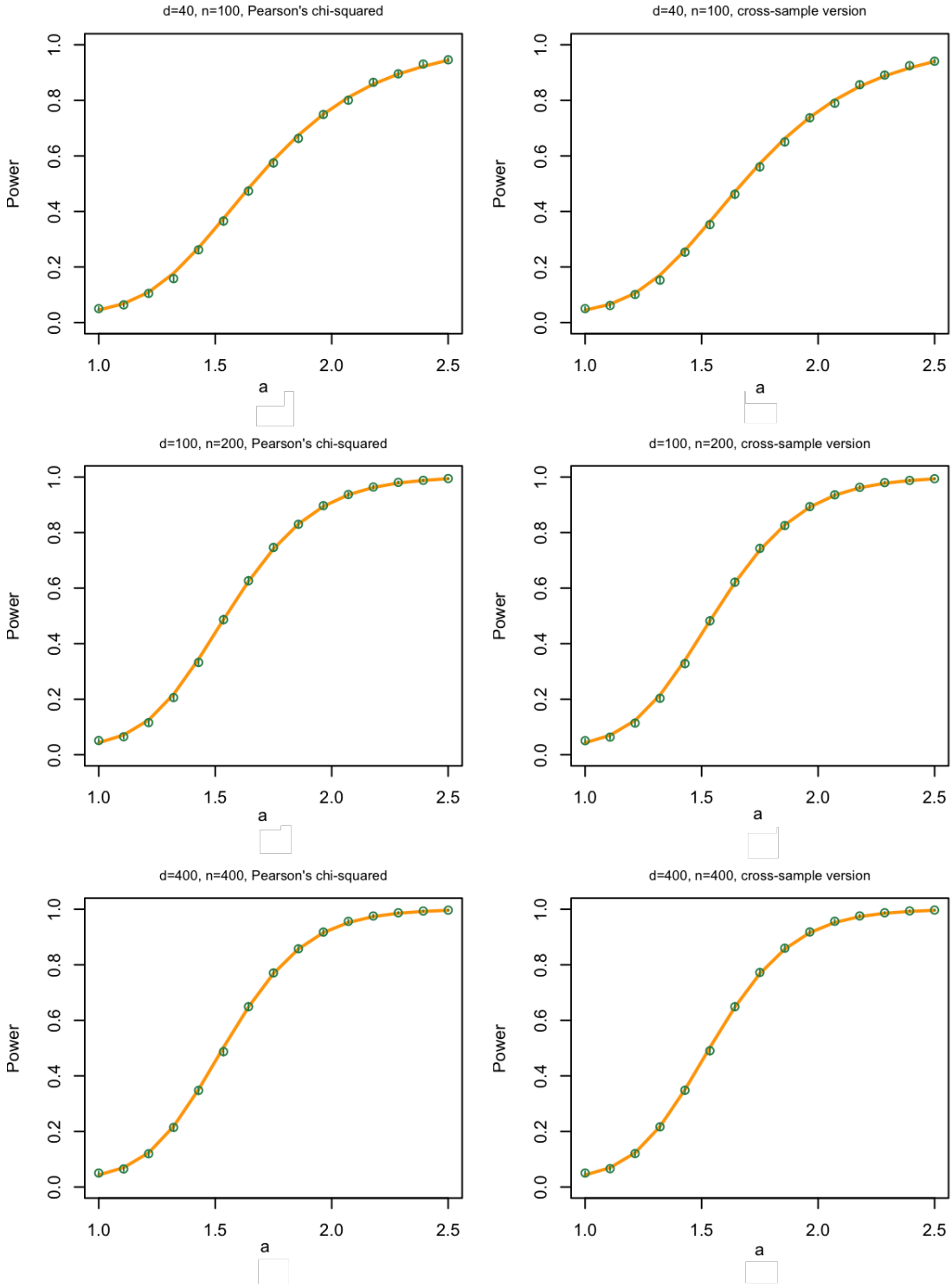


Figure 3: The orange solid line depicts the change of theoretical power with the deviation from the null probability vector. The black circles show the scatterings of the empirical powers with the error bars representing minus and plus three standard deviation.

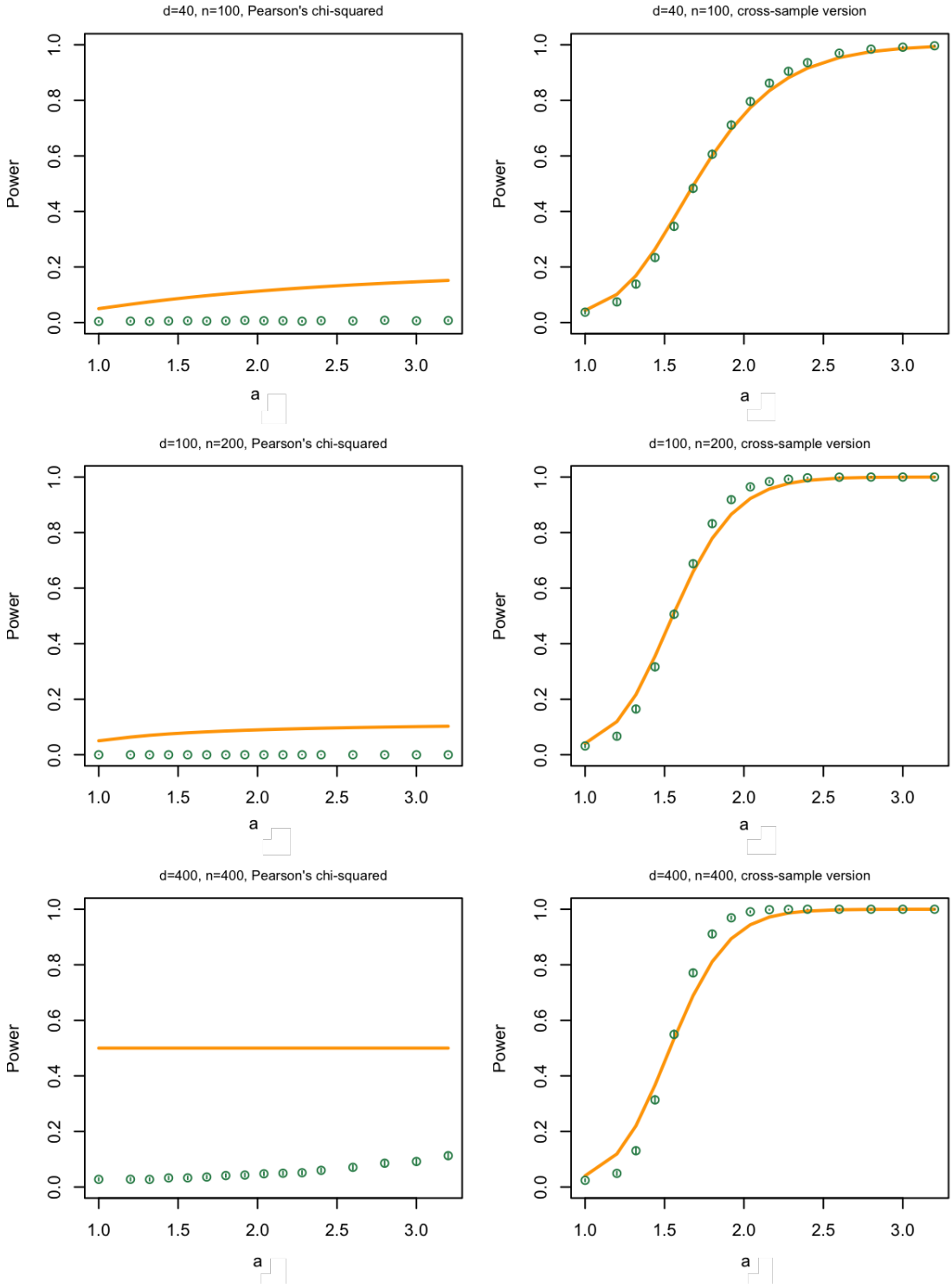


Figure 4: The change of theoretical powers along the deviation from the null is depicted by the orange solid line. The black circles show the scattering of empirical powers with the error bar representing minus and plus three standard deviation.

The conclusions are asymptotic and should be interpreted with their theorem-specific assumptions. Convergence in the supremum distance between distribution functions controls absolute errors in rejection probabilities, but does not by itself provide relative tail accuracy when significance levels tend to zero. Likewise, an asymptotic occupancy condition does not supply a universal finite-sample cutoff for deciding whether an approximation is adequate. Quantitative error bounds would help determine how approximation accuracy depends on the cell probability array and the distance from the validity boundary. Such bounds could also guide the use of normal and adjusted chi-squared references at moderate sample sizes.

Several further questions arise from this framework. Characterizing non-Gaussian limits when the approximation criteria fail would provide a fuller description of the regimes outside the validity boundaries. Extending the analysis to more general fitted multinomial models, especially when the number of estimated parameters increases, would clarify how estimation changes the centering, variance, and approximation conditions. Data-dependent partitions and dependent observations present additional questions because they alter the multinomial structure used in the current proofs. These extensions would require accounting explicitly for the fluctuations introduced by fitting, selection, or dependence.

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